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Active resistance control: a contact control method for detumbling non-cooperative spacecraft by a robotic arm

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Abstract Non-cooperative spacecraft often exhibit a tumbling motion, posing significant challenges for successful capture by space robots. Prior to capturing attempts, detumbling operations have become essential. Among various detumbling technologies, the robotic arm-based detumbling approach is relatively mature and easy to implement. However, key issues, particularly maintaining contact control problems, still need to be addressed. This paper provides a detailed investigation of the contact control issue and proposes a new contact control approach called active resistance control (ARC). Theoretical proof and numerical simulation show that this method can prevent the post-collision separation and track the desired contact force. Notably, the theoretical validation of ARC is derived from generalized contact model expressions rather than a specific contact model, ensuring broader applicability. Unlike existing methods, the control law of ARC relies only on measured contact force and does not require prior knowledge of the target's inertial or contact parameters. This eliminates the need for target parameter identification in real-world tasks, making ARC more practical to implement. To apply this method to the detumbling tasks, this study introduces an ARC-based contact control strategy. The

simulation results validate the efficiency of the proposed contact control strategy across diverse detumbling scenarios.

Keywords Detumbling · Non-cooperative spacecraft · Space robot · Contact control · Active resistance control

1 Introduction

Since the beginning of the twenty-first century, the number of decommissioned or malfunctioning spacecraft has been increasing daily, owing to the rapid growth in the number of spacecraft launched. These spacecraft waste valuable space orbit resources and threaten the safety of spacecraft on orbit [1–3]. In response to this challenge, the concept of on-orbit servicing has emerged as a hot research focus in the field of space exploration [4–6]. Capturing the target for servicing is a prerequisite for OOS, but it becomes particularly challenging when dealing with tumbling non-cooperative spacecraft. Prior to capturing attempts, detumbling operations have become essential [7, 8]. Despite many researchers conducted by scholars over the past decade on the detumbling problem, numerous challenges remain unresolved, which will be addressed later.

For detumbling tasks, the operation process consists of three phases: far-range rendezvous, close-

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range rendezvous and detumbling. The far-range rendezvous phase refers to the process in which the chaser spacecraft gradually approaches the target from an initial distance of several kilometers. During this phase, the main objective is to reduce the relative distance between the chaser spacecraft and the target. The close-range rendezvous phase begins when the chaser enters the proximity of the target within a few tens of meters. During this stage, the position error between the chaser and the target is further reduced. In addition, attitude synchronization and collision avoidance must also be performed at this stage to ensure the success of the subsequent detumbling operation. A detailed discussion of the above two stages can be found in Refs. [9–12].

This paper primarily focuses on the detumbling phase, during which a detumbling torque is applied to the target to gradually reduce its angular velocity. Currently, the existing detumbling strategies can be divided into two categories: contactless strategies and contact strategies. Contactless detumbling strategies utilize a range of techniques, including laser ablation, plume impingement, eddy currents, and electrostatic forces, to effectively mitigate the rotational motion of the target object [13–23]. On the other hand, contact strategies involve utilizing elastic and friction forces generated through physical contact to eliminate the rotation of the target. In comparison, contact detumbling strategies are better suited for the detumbling task of non-cooperative spacecraft as they can generate higher detumbling torques [24–26]. However, in contact detumbling strategies, a collision between the contactor and the target is highly likely to cause immediate separation between the two, making stable contact cannot be formed. As a result, some scholars have proposed detumbling strategies based on multiple tentative contacts, wherein the robotic arm repeatedly collides with the target to generate detumbling torques [27–30]. Nevertheless, the torque produced by such repeated contacts is discontinuous, which is inefficient in achieving effective detumbling. Moreover, repeated impacts increase the risk of structural damage to both the robotic arm and the target.

To overcome the above shortcomings, scholars have proposed various solutions to prolong post-collision contact, thereby extending the action time of the detumbling torque. For instance, some researchers have utilized flexible mechanisms to achieve

stable contact after a collision. Nishida and Kawamoto [31] employed a brush-type contactor to mitigate the tumbling motion of space debris. Similarly, Sun et al. [32] utilized flexible brushes to detumble the target and proposed the H_∞ optimal control to ensure global robustness and stability. Ma et al. [33] analyzed the dynamics of the detumbling process of brush contactors based on flexible multi-body dynamics. In Ref. [34], the authors adopted the deep reinforcement learning method to design a detumbling control strategy. Apart from flexible brushes, another viable option for a contactor in detumbling operations is a flexible bar. In our previous work [35, 36], we examined the feasibility of detumbling operations using flexible bars, with the body of the spacecraft selected as the target for contact. Similarly, in Refs. [25] and [26] the authors investigated the feasibility of detumbling operations based on flexible bars by selecting solar panels of a non-cooperative spacecraft as the objects to be contacted.

Although flexible mechanisms can attenuate rotation by applying external torque to the target, they still have some shortcomings. The biggest one is that the flexible mechanisms are difficult to control since they have underactuated characteristics. Compared to flexible contactors, rigid contactors obviously have better control performance. However, rigid contactors cannot realize continuous contact via material deformation, so active control is required to prolong the contact duration. For instance, Uyama et al. [37, 38] and Chen et al. [24] have each proposed impedance-based control approaches, tailored for single-manipulator and dual-manipulator detumbling tasks, respectively. Wang et al. [39] used a hybrid position/force control to solve the contact control problem in detumbling tasks. Additionally, our group has introduced a control approach, namely overdamping control, which can be employed to prolong the contact duration during robotic-arm-based detumbling [40]. Importantly, the issue of prolonging post-collision contact also arises in capture tasks and has been extensively investigated in the context of non-cooperative target capture. For instance, Yoshida et al. [41, 42] introduced an impedance control framework that adjusts the manipulator's virtual mass to match that of the target, enabling stable contact and minimizing rebound during collisions. Stolfi et al. [43] combined PD and impedance control in a dual-arm system to reduce impact forces and prolong contact

with flexible targets. Furthermore, Wu et al. [44] developed a resolved motion admittance control method to improve the compliance of space robots. Based on the disturbance observers, Flores-Abad et al. [45, 46] designed a force-sensor-less impedance controller, allowing compliant capture without direct force measurements. Apart from the above-mentioned impedance-control-based methods, some researchers have also developed contact control strategies based on hybrid force/position control or sliding mode control. For example, Shi et al. [47] proposed a hybrid motion/force control strategy for free-floating space robots, which enables compliant interaction with targets under complex multi-DOF contact geometries. Han et al. [48] developed a compliant control scheme combined with adaptive sliding mode control, enabling dual-arm robots to capture and detumble non-cooperative targets without a specific gripping point. Wang et al. [49] further enhanced the controller's robustness by employing a smooth quasi-continuous second-order sliding mode controller (SQC2S) to reduce chattering and ensure accurate contact force tracking in 3-D space. In addition to the above studies, our previous works have proposed damping control [50, 51] to achieve soft contact between the satellite and the target.

The research conducted in the aforementioned works has demonstrated the feasibility of rigid-contact-based detumbling strategies, but has also revealed certain limitations. Firstly, these methods rely on prior knowledge of the target's inertial or contact parameters, which can be challenging to measure. Secondly, existing control methods mainly focus on prolonging the contact duration rather than achieving stable contact after a collision, which restricts the operating time for exerting detumbling torques. Thirdly, many of these methods lack control over the magnitude of the contact force, resulting in uncontrolled detumbling torques. To overcome the aforementioned drawbacks, we previously developed a contact control approach, namely resistance control (RC), which can establish stable contact [52]. However, as a feedforward control method, this method relies on many prior parameters, including the coefficient of restitution, contact stiffness, target mass, and initial relative velocity. Acquiring these parameters with precision, particularly in the context of non-cooperative target detumbling tasks, presents a significant challenge. Furthermore, the RC method

amplifies the impact generated by collisions, thereby increasing the risk of damage to the robotic arm or target. In this paper, we addressed the limitations of RC by proposing a new one-dimensional contact control approach, namely the active resistance control (ARC) method. This method, unlike the RC method, adopts a feedback control approach that relies solely on the measured contact force. Thus, this approach eliminates the need for identifying the target's inertia and contact parameters in real-world applications, thereby making ARC more practical to implement. Additionally, ARC allows for maintaining contact and controlling the contact force. To prove the ARC method, we initially simplified the one-dimensional collision problem to a two-ball collision problem. Utilizing the two-ball collision model, we established differential equations for the penetration depth of contact (Δx), thereby demonstrating the theoretical efficacy of the ARC method. Subsequently, for the application of the ARC method to the detumbling task (Fig. 1), we introduced an ARC-based contact control strategy, which can be regarded as a hybrid force/position control strategy. The simulation results validated the effectiveness and robustness of the proposed strategy across various scenarios. Notably, simulation results showed that the ARC-based contact control strategy accomplishes nearly equivalent detumbling effects to the RC-based contact control strategy, yet with reduced collision impact.

The structure of the paper is as follows: Sect. 2 presents the dynamic model of the space robot. Section 3 describes the Active Resistance Control (ARC) method along with the proposed contact control strategy. Numerical simulations are provided in Sect. 4, and finally, Sect. 5 concludes the paper.

2 Dynamic models

This paper utilizes a space robot to detumble a non-cooperative target, as illustrated in Fig. 1. Figure 2 presents the schematic representation of the space robot's mechanism along with the target. The space robot consists of a base (body B_1), a robotic arm (bodies B_2 – B_5) and a contactor (body B_6). The robotic arm's joints are denoted as H_i , where H_1 – H_6 represent rotational joints, and H_7 is a translational joint. The $\sum -xyz$ frame serves as the inertial reference frame.

Fig. 1 Robotic-arm-based detumbling tasks

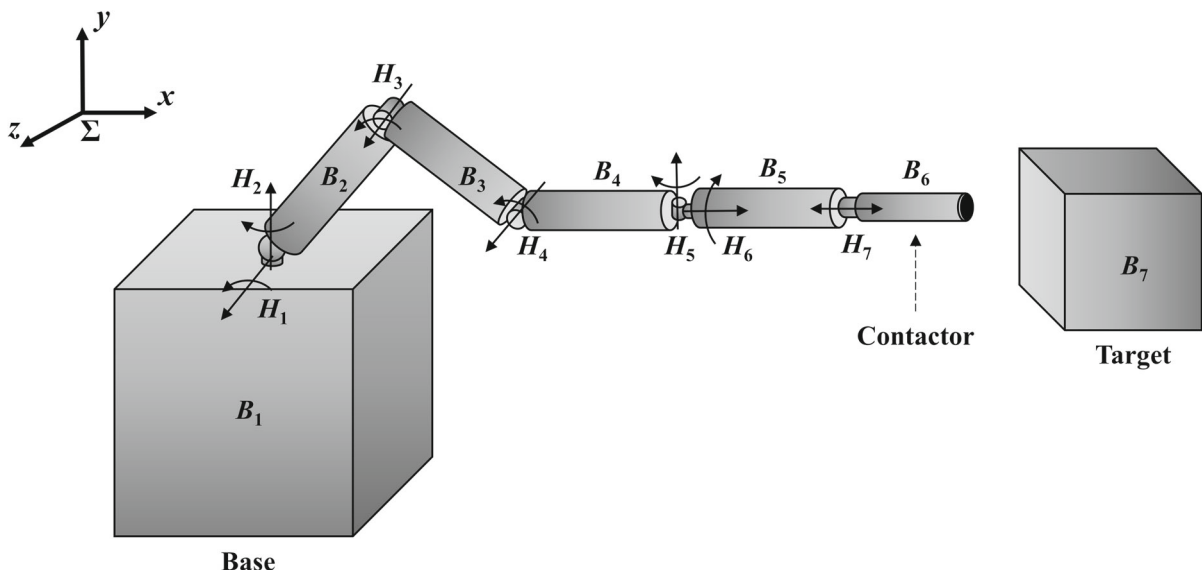
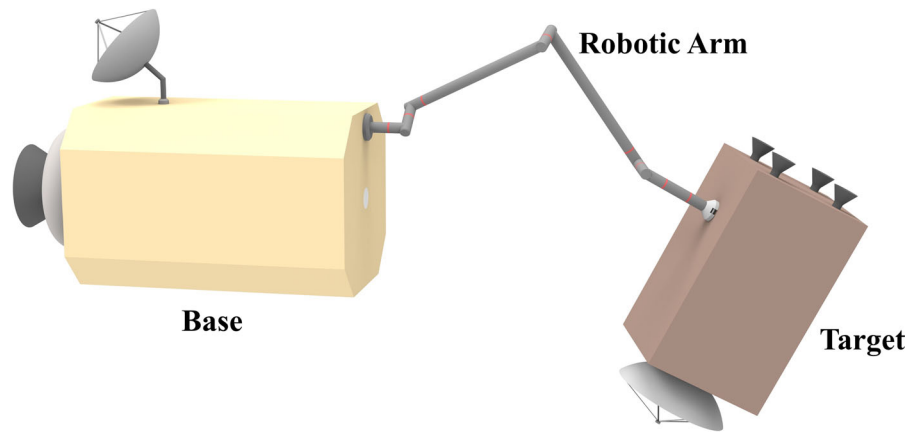


Fig. 2 Space robot and non-cooperative target

The robot's dynamic equation can be expressed as [53]

$$-\mathbf{M}\ddot{\mathbf{q}} - \mathbf{c} + \mathbf{f} + \mathbf{u} = \mathbf{0} \quad (1)$$

where $\mathbf{q} = [\mathbf{q}_{\text{base}}^T, \mathbf{q}_{\text{arm}}^T, q_{\text{contactor}}]^T \in \mathbb{R}^{13 \times 1}$ represents the generalized coordinates. Specifically, $\mathbf{q}_{\text{base}} = [x, y, z, \alpha, \beta, \gamma]^T$ denotes the base's position and attitude, $\mathbf{q}_{\text{arm}} = [\theta_1, \theta_2, \dots, \theta_6]^T$ denotes the robotic arm's joint angles, and $q_{\text{contactor}}$ representing the displacement of the contactor relative to the end of the robot; The generalized mass matrix is denoted as $\mathbf{M}$, while $\mathbf{c}$ and $\mathbf{f}$ represent the generalized inertia force and the generalized external force, respectively. The

control input is defined as $\mathbf{u} = [\mathbf{u}_{\text{base}}^T, \mathbf{u}_{\text{arm}}^T, u_{\text{contactor}}]^T$, where $\mathbf{u}_{\text{base}}^T \in \mathbb{R}^{6 \times 1}$ represents the control forces and torques acting on the base, $\mathbf{u}_{\text{arm}}^T \in \mathbb{R}^{6 \times 1}$ represents the robotic arm's joint driving torques, and $u_{\text{contactor}}$ representing the joint driving force. Now, the dynamic model for the system has been formulated, with detailed information available in Ref. [53].

3 Detumbling control

In the rigid-contactor-based detumbling task, the detumbling controller faces the challenge of

establishing stable contact between the robotic arm and the target. To address this challenge, Uyama et al. divided the contact control in detumbling into a one-dimension contact control along the normal direction of the contact surface and a motion-tracking control along the other directions [37]. The primary goal of the former is to regulate the relative motion between the contactor and the target along the direction of the contact force, whereas the latter is focused on directing the contactor along the intended trajectory in the other dimensions. While the existing control methods have shown satisfactory performance, they still have certain limitations. One drawback is their reliance on prior knowledge of the target's inertial parameters and contact parameters for controller design, which is challenging to measure accurately. Furthermore, these methods primarily focus on extending the contact time after a collision and do not provide effective control over the magnitude of the contact force, which is a significant shortcoming of contact-type detumbling methods compared to contactless-type detumbling methods.

To address the aforementioned limitations, this study proposes a novel contact control method called ARC. This approach enables the formation of stable contact between two objects following a collision and provides precise regulation of the contact force, all without requiring information on the target's inertial and contact parameters. The following subsection provides more details of ARC, followed by the presentation of a contact control strategy based on ARC.

3.1 The ARC method for contact control

As previously discussed, the contact control problem in detumbling can be divided into a one-dimensional contact control problem and a motion-tracking control

problem. In this study, the one-dimensional contact control problem is simplified into a two-ball contact control problem, as illustrated in Fig. 3, where S_1 and S_2 denote the target spacecraft and the contactor. Using Newton's laws of motion, the dynamics equations of the two-ball collision are:

$$\begin{cases} m_1 \ddot{x}_1 = -F_c \\ m_2 \ddot{x}_2 = F_c - u \end{cases} \quad (2)$$

where m_1 and m_2 are the masses of S_1 and S_2 , x_1 and x_2 denote the positions of S_1 and S_2 , F_c is the contact force, and u is the control force. The contact depth Δx is defined as:

$$\Delta x = (x_1 + r_1) - (x_2 - r_2) \quad (3)$$

where r_1 and r_2 denote the radius of S_1 and S_2 ; $\Delta x \geq 0$ indicates contact and $\Delta x < 0$ indicates separation. According to Eqs. (2) and (3), we can obtain the differential equation for the contact depth Δx :

$$\Delta \ddot{x} = -\lambda F_c + \frac{u}{m_2} \quad (4)$$

Where $\lambda = (m_1 + m_2)/m_1 m_2$ is the equivalent mass ratio.

In this study, a novel control method, referred to as the active resistance control (ARC), is proposed to prevent post-collision separation and track the desired contact force. The control law of ARC is as follows:

$$u(k) = \begin{cases} \beta_1 \max(F_c) & \text{if } 0 \leq k\Delta t \leq t_s \\ u(k-1) + \beta_2 e(k-1) & \text{if } k\Delta t > t_s \end{cases} \quad (5)$$

where $\max(F_c)$ represents the maximum contact force from the initial moment to the current moment; $u(k)$ denotes the control force at k -th sampling instant; $e(k-1) = F_c^*(k-1) - F_c(k-1)$, where $F_c^*(k-1)$ and $F_c(k-1)$ denote the desired and actual contact force at $(k-1)$ -th sampling instant, respectively; Δt is the sampling period; t_s is the switching moment. The ranges of the control parameters β_1 and β_2 are:

$$\begin{aligned} \beta_1 &\geq 0.5 \\ 0 < \beta_2 < 1 \end{aligned} \quad (6)$$

As shown in Eq. (5), the control process of ARC is divided into two phases. The first phase ($0 \leq k\Delta t \leq t_s$) is to prevent post-collision separation; the second phase ($k\Delta t > t_s$) is to track the desired contact force. Once a stable contact forms, i.e., contact force

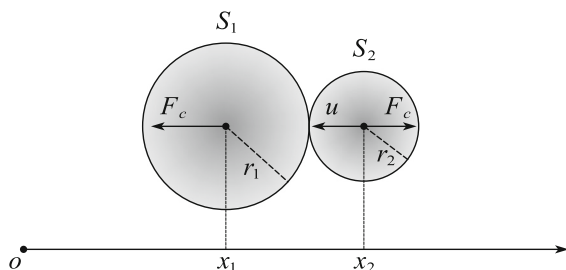


Fig. 3 Two-ball collision model

stabilization, the control switches from the first phase to the second phase. Next section will present the theory proof of the effectiveness of ARC.

3.2 Proof of ARC's effectiveness

In this section, the effectiveness of ARC is demonstrated based on the following assumptions:

Assumptions 1 $m_1 \gg m_2$, where m_1 and m_2 are the masses of the target spacecraft and the contactor, respectively.

Assumptions 2 The desired contact force $F_c^*(t)$ satisfies the Lipschitz condition, i.e., for any $t_1 > 0$ and $t_2 > 0$, there exists a constant L such that:

$$|F_c^*(t_1) - F_c^*(t_2)| \leq L|t_1 - t_2| \quad (7)$$

Assumptions 3 External orbital disturbances, such as atmospheric drag, solar radiation pressure, and J_2 perturbation, can be ignored in this study.

3.2.1 The effectiveness of ARC in preventing post-collision separation

According to the classical collision theory, a collision process can be divided into the compression phases and the restitution phases (the compression phase is defined as the phase where the contact depth Δx increases, whereas the restitution phase is defined as the phase where Δx decreases). A complete compression-restitution phase is shown in Fig. 4. In this figure, $\Delta x = \Delta x_{\max}$ and $\Delta x = \Delta x_p$ respectively denote the starting and ending point of the restitution phase, and $\Delta x = \Delta x_p$ is the permanent deformation after collision.

Note that $\Delta x_p = 0$ for the viscoelastic contact model and $\Delta x_p > 0$ for the elastoplastic contact model [54, 55]. Observing Fig. 4, it can be seen that if separation takes place after a collision, then the separation must occur in the restitution phase. Therefore, to prove the effectiveness of ARC in preventing separation, it is sufficient to demonstrate that separation does not occur during the restitution phase. Next, we will present the proof.

Substituting the control force of the first phase in (5) into (4) yields:

$$\Delta \ddot{x} = -\lambda F_c + \frac{\beta_1 \max(F_c)}{m_2} \quad (8)$$

This equation can be viewed as the motion equation of an object with a mass of 1 kg under the external force $-\lambda F_c$ and $\beta_1 \max(F_c)/m_2$. Let the work done by $-\lambda F_c$ and $\beta_1 \max(F_c)/m_2$ in the restitution phase ($\Delta x_p \leq \Delta x \leq \Delta x_{\max}$) be W_1 and W_2 , and T is the kinetic energy at $\Delta x = \Delta x_p$. According to the definition of work and kinetic energy, we can get that:

$$W_1 = \int_{\Delta x_{\max}}^{\Delta x_p} (-\lambda F_c) d\Delta x = \int_{\Delta x_p}^{\Delta x_{\max}} \lambda F_c d\Delta x \quad (9)$$

$$\begin{aligned} W_2 &= \int_{\Delta x_{\max}}^{\Delta x_p} \frac{\beta_1 \max(F_c)}{m_2} d\Delta x \\ &= - \int_{\Delta x_p}^{\Delta x_{\max}} \frac{\beta_1 \max(F_c)}{m_2} d\Delta x \end{aligned} \quad (10)$$

$$T = \frac{1}{2} \left(\Delta \dot{x}|_{\Delta x = \Delta x_p} \right)^2 \quad (11)$$

where $\Delta \dot{x}|_{\Delta x = \Delta x_p}$ denotes the contact velocity at $\Delta x = \Delta x_p$. Since $\Delta \dot{x} = 0$ at $\Delta x = \Delta x_{\max}$, according to

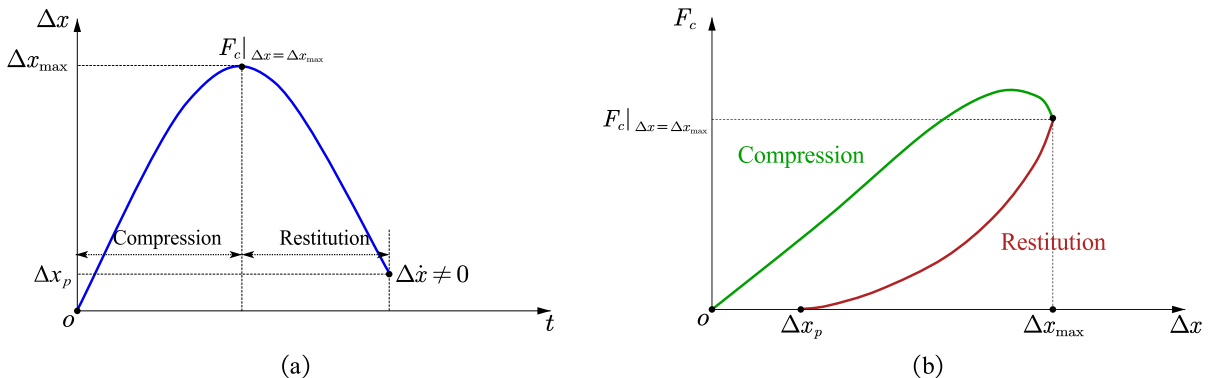


Fig. 4 **a** The curve of Δx vs time; **b** The curve of F_c vs Δx

the law of conservation of energy, the relation of them is:

$$W_1 + W_2 = T \quad (12)$$

If separation occurs, then $\Delta \dot{x}|_{\Delta x = \Delta x_p} > 0$, meaning $T > 0$. Accordingly, a necessary condition for separation is $W_1 + W_2 > 0$, i.e.,

$$\text{separation} \Rightarrow W_1 + W_2 > 0 \quad (13)$$

Considering the contrapositive of above proposition, a sufficient condition for non-separation is $W_1 + W_2 \leq 0$, i.e.,

$$W_1 + W_2 \leq 0 \Rightarrow \text{non - separation} \quad (14)$$

Substituting (9) and (10) into (14) yields:

$$\int_{\Delta x_p}^{\Delta x_{\max}} \lambda F_c d\Delta x \leq \int_{\Delta x_p}^{\Delta x_{\max}} \frac{\beta_1 \max(F_c)}{m_2} d\Delta x \Rightarrow \text{non - separation} \quad (15)$$

According to the Assumption 1, we can get that:

$$\lambda = \frac{m_1 + m_2}{m_1 m_2} \approx \frac{m_1}{m_1 m_2} = \frac{1}{m_2} \quad (16)$$

Substituting (16) into (15) yields:

$$\int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x \leq \beta_1 \int_{\Delta x_p}^{\Delta x_{\max}} \max(F_c) d\Delta x \Rightarrow \text{non - separation} \quad (17)$$

Let $F_c|_{\Delta x = \Delta x_{\max}}$ denote the contact force at $\Delta x = \Delta x_{\max}$ (Fig. 4). Since is $\Delta x = \Delta x_{\max}$ the starting point of the restitution phase, according to the definition of $\max(F_c)$, we have:

$$\max(F_c) \geq F_c|_{\Delta x = \Delta x_{\max}} \text{ in restitution phase} \quad (18)$$

Combining (18) and $\beta_1 \geq 0.5$, we can get the lower bound of $\beta_1 \int_{\Delta x_p}^{\Delta x_{\max}} \max(F_c) d\Delta x$:

$$\begin{aligned} \beta_1 \int_{\Delta x_p}^{\Delta x_{\max}} \max(F_c) d\Delta x &\geq \frac{1}{2} \int_{\Delta x_p}^{\Delta x_{\max}} F_c|_{\Delta x = \Delta x_{\max}} d\Delta x \\ &= \frac{F_c|_{\Delta x = \Delta x_{\max}} (\Delta x_{\max} - \Delta x_p)}{2} \end{aligned} \quad (19)$$

According to (19), we can get that:

$$\begin{aligned} \int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x &\leq \frac{F_c|_{\Delta x = \Delta x_{\max}} (\Delta x_{\max} - \Delta x_p)}{2} \\ \Rightarrow \int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x &\leq \beta_1 \int_{\Delta x_p}^{\Delta x_{\max}} \max(F_c) d\Delta x \end{aligned} \quad (20)$$

Combining (17) and (20), we have:

$$\begin{aligned} \int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x &\leq \frac{F_c|_{\Delta x = \Delta x_{\max}} (\Delta x_{\max} - \Delta x_p)}{2} \\ \Rightarrow \text{non - separation} \end{aligned} \quad (21)$$

The above equation indicates the sufficient condition for non-separation is $\int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x \leq (F_c|_{\Delta x = \Delta x_{\max}} (\Delta x_{\max} - \Delta x_p))/2$. Next, we will prove that this equation holds for both elastoplastic and viscoelastic contact models, thereby proving the effectiveness of ARC.

3.2.1.1 Case I: elastoplastic contact model In most elastoplastic models [54, 56–58], the contact force during the restitution phase ($\Delta x_p \leq \Delta x \leq \Delta x_{\max}$) can be expressed in the following form:

$$F_c = c_1 \left(\frac{\Delta x - \Delta x_p}{c_2} \right)^\alpha \quad (22)$$

where c_1 and c_2 are positive constants and $\alpha \geq 1$. Taking the second derivative of (22) yields:

$$\frac{d^2 F_c}{d\Delta x^2} = \frac{\alpha(\alpha - 1)c_1}{c_2^2} \left(\frac{\Delta x - \Delta x_p}{c_2} \right)^{\alpha-2} \geq 0 \quad (23)$$

Equation (23) indicates that the contact force curve with respect to the contact depth is convex during the restitution phase. In Fig. 5, Curve I represents the contact force curve in the restitution phase, while Curve II connects two points $(\Delta x_p, 0)$ and

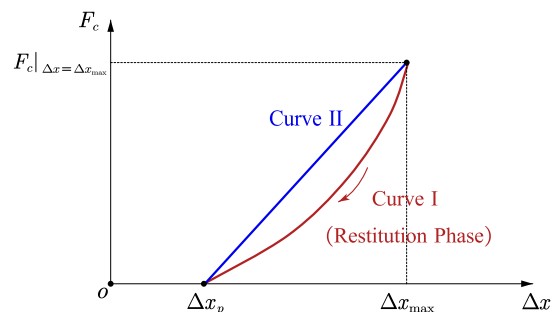


Fig. 5 The curve of contact force versus contact depth in the restitution phase

$(\Delta x_{\max}, F_c|_{\Delta x=\Delta x_{\max}})$. The area under Curve I is $\int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x$, and the area under Curve II is $(F_c|_{\Delta x=\Delta x_{\max}}(\Delta x_{\max} - \Delta x_p))/2$. Since Curve I is convex, Curve I is always below Curve II. So, $\int_{\Delta x_p}^{\Delta x_{\max}} F_c d\Delta x \leq (F_c|_{\Delta x=\Delta x_{\max}}(\Delta x_{\max} - \Delta x_p))/2$.

According to (21), we can get that ARC is capable of preventing separation when the contact force is modeled using elastoplastic contact models.

3.2.1.2 Case II: viscoelastic contact model Most of viscoelastic contact model can be expressed in the following form [55, 59, 60]:

$$F_c = k_c \Delta x^\alpha + \xi(\cdot) \Delta \dot{x} \quad (24)$$

where Δx and $\Delta \dot{x}$ denote the contact depth and the contact velocity, respectively ($\Delta x \geq 0$ represents contact, $\Delta x < 0$ represents separation); k_c is the contact stiffness; the exponent α is a constant and the equivalent damping $\xi(\cdot)$ is a function of Δx . They satisfy:

$$\begin{cases} \alpha \geq 1 \\ \xi(\cdot) \geq 0 \end{cases} \quad (25)$$

Note that the permanent deformation is $\Delta x_p = 0$ in the viscoelastic contact model, so (21) is written as:

$$\begin{aligned} \int_0^{\Delta x_{\max}} F_c d\Delta x &\leq \frac{F_c|_{\Delta x=\Delta x_{\max}} \cdot \Delta x_{\max}}{2} \\ \Rightarrow \text{non - separation} \end{aligned} \quad (26)$$

In the restitution phase, considering that $\Delta \dot{x} \leq 0$ and $\xi(\cdot) \geq 0$, we can get $\xi(\cdot) \Delta \dot{x} \leq 0$. Thus, contact force in the restitution phase is:

$$F_c = k_c \Delta x^\alpha + \xi(\cdot) \Delta \dot{x} \leq k_c \Delta x^\alpha \quad (27)$$

According to (27), we can calculate the upper bound of $\int_0^{\Delta x_{\max}} F_c d\Delta x$:

$$\int_0^{\Delta x_{\max}} F_c d\Delta x \leq \int_0^{\Delta x_{\max}} k_c \Delta x^\alpha d\Delta x = \frac{k_c \Delta x_{\max}^{\alpha+1}}{\alpha+1} \quad (28)$$

Furthermore, since $\Delta \dot{x} = 0$ at $\Delta x = \Delta x_{\max}$, according to (24), we can get that $F_c|_{\Delta x=\Delta x_{\max}} = k_c \Delta x_{\max}^\alpha$. So,

$$\frac{F_c|_{\Delta x=\Delta x_{\max}} \cdot \Delta x_{\max}}{2} = \frac{k_c \Delta x_{\max}^{\alpha+1}}{2} \quad (29)$$

According to (25), $\alpha \geq 1$. Combining with (29), we have:

$$\frac{F_c|_{\Delta x=\Delta x_{\max}} \cdot \Delta x_{\max}}{2} \geq \frac{k_c \Delta x_{\max}^{\alpha+1}}{\alpha+1} \quad (30)$$

Combining (28) and (30), we can get that $\int_0^{\Delta x_{\max}} F_c d\Delta x \leq (F_c|_{\Delta x=\Delta x_{\max}} \cdot \Delta x_{\max})/2$. According to (26), ARC can prevent separation when the contact force is described by viscoelastic contact models. In summary, we can conclude that ARC is effective for preventing post-collision separation.

3.2.2 The effectiveness of ARC in tracking desired contact force

Contact theory tells us that the collision process is always accompanied by energy dissipation, which may be attributed to viscous damping or plastic deformation [61]. The dissipated energy is the initial kinetic energy of the system. If no separation occurs after the collision, the initial relative kinetic energy of the colliding bodies must be fully dissipated. In this case, there will be no relative motion between the colliding bodies, i.e., $\Delta \ddot{x} = 0$ and $\Delta \dot{x} = 0$. Substituting $\Delta \ddot{x} = 0$ into (4), we can obtain the contact force when stable contact forming is:

$$F_c = \frac{1}{\lambda m_2} u \quad (31)$$

The discrete expression of (31) is:

$$F_c(k) = \frac{1}{\lambda m_2} u(k) \quad (32)$$

According to (5), the control law in the second phase is:

$$u(k) = u(k-1) + \beta_2 e(k-1) \quad (33)$$

where $u(k)$ and $u(k-1)$ denote the control force at k -th and $(k-1)$ -th sampling instant; $e(k-1) = F_c^*(k-1) - F_c(k-1)$, where $F_c^*(k-1)$ and $F_c(k-1)$ denote the desired and actual contact force at $(k-1)$ -th sampling instant, respectively; the range of the control

parameter β_2 is $0 < \beta_2 < 1$. Next, we will prove the ARC's effectiveness for tracking desired contact force in two cases: (I) $F_c^*(k)$ is constant; (II) $F_c^*(k)$ is time-varying.

Case I: $F_c^*(k)$ is constant, i.e. $F_c^*(k) = F_c^* = \text{const.}$

As the desired contact force is a constant, the control error at k -th sampling instant is:

$$|e(k)| = |F_c^*(k) - F_c(k)| = |F_c^* - F_c(k)| \quad (34)$$

Substituting (32) and (33) into (34) yields:

$$\begin{aligned} |e(k)| &= \left| F_c^* - \frac{1}{\lambda m_2} u(k) \right| = \left| F_c^* - \frac{1}{\lambda m_2} (u(k-1) + \beta_2 e(k-1)) \right| \\ &= \left| F_c^* - \frac{1}{\lambda m_2} u(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| = \left| F_c^* - F_c(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| e(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| 1 - \frac{\beta_2}{\lambda m_2} \right| |e(k-1)| \end{aligned} \quad (35)$$

Equation (35) shows that

$$|e(k)| = \left| 1 - \frac{\beta_2}{\lambda m_2} \right| |e(k-1)|, \text{ which follows:}$$

$$\begin{aligned} |e(k)| &= \left| 1 - \frac{\beta_2}{\lambda m_2} \right| |e(k-1)| = \left| 1 - \frac{\beta_2}{\lambda m_2} \right|^2 |e(k-2)| \\ &= \dots = \left| 1 - \frac{\beta_2}{\lambda m_2} \right|^{k-1} |e(1)| \end{aligned} \quad (36)$$

Since $0 < \beta_2 < 1$ and $\lambda m_2 = (m_1 + m_2)/m_1 > 1$, we can get that:

$$0 < \left(1 - \frac{\beta_2}{\lambda m_2} \right) < 1 \quad (37)$$

Substituting (37) into (36) and taking the limit of $|e(k)|$ yields:

$$\lim_{k \rightarrow \infty} |e(k)| = \lim_{k \rightarrow \infty} \left(\left| 1 - \frac{\beta_2}{\lambda m_2} \right|^{k-1} |e(1)| \right) = 0 \quad (38)$$

Therefore, we can obtain that tracking error converges to zero as $F_c^*(k)$ is constant.

Case II: $F_c^*(k)$ is time-varying.

As the desired contact force $F_c^*(k)$ is time-varying, the control error at k -th sampling instant is:

$$|e(k)| = |F_c^*(k) - F_c(k)| \quad (39)$$

Substituting (32) and (33) into (39) yields:

$$\begin{aligned} |e(k)| &= \left| F_c^*(k) - \frac{1}{\lambda m_2} u(k) \right| \\ &= \left| F_c^*(k) - \frac{1}{\lambda m_2} (u(k-1) + \beta_2 e(k-1)) \right| \\ &= \left| F_c^*(k) - \frac{1}{\lambda m_2} u(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| F_c^*(k) - F_c(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| F_c^*(k) - F_c^*(k-1) + F_c^*(k-1) - F_c(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| F_c^*(k) - F_c^*(k-1) + e(k-1) - \frac{\beta_2}{\lambda m_2} e(k-1) \right| \\ &= \left| F_c^*(k) - F_c^*(k-1) + \left(1 - \frac{\beta_2}{\lambda m_2} \right) e(k-1) \right| \\ &\leq |F_c^*(k) - F_c^*(k-1)| + \left(1 - \frac{\beta_2}{\lambda m_2} \right) |e(k-1)| \end{aligned} \quad (40)$$

According to the Assumption 2, $F_c^*(k)$ satisfies the Lipschitz condition. So, there exists a positive constant L such that:

$$|F_c^*(k) - F_c^*(k-1)| \leq L(k\Delta t - (k-1)\Delta t) = L\Delta t \quad (41)$$

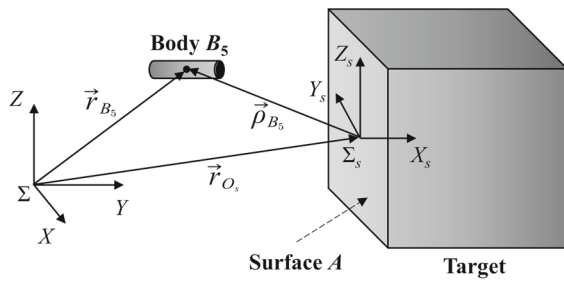
Substituting (41) into (40) yields:

$$|e(k)| \leq L\Delta t + \left(1 - \frac{\beta_2}{\lambda m_2} \right) |e(k-1)| \quad (42)$$

According to (42), the following recurrence relation can be obtained:

$$\begin{aligned} |e(2)| &\leq L\Delta t + \left(1 - \frac{\beta_2}{\lambda m_2} \right) |e(1)| \\ |e(3)| &\leq L\Delta t + \left(1 - \frac{\beta_2}{\lambda m_2} \right) |e(2)| \leq L\Delta t \left(1 + \left(1 - \frac{\beta_2}{\lambda m_2} \right) \right) \\ &\quad + \left(1 - \frac{\beta_2}{\lambda m_2} \right)^2 |e(1)| \\ &\vdots \\ |e(k)| &\leq L\Delta t \left(1 + \left(1 - \frac{\beta_2}{\lambda m_2} \right) + \dots + \left(1 - \frac{\beta_2}{\lambda m_2} \right)^{k-2} \right) \\ &\quad + \left(1 - \frac{\beta_2}{\lambda m_2} \right)^{k-1} |e(1)| \end{aligned} \quad (43)$$

According to (37) and (43), taking the limit of $|e(k)|$ yields:



$$\begin{aligned}
 \lim_{k \rightarrow \infty} |e(k)| &\leq \lim_{k \rightarrow \infty} \left(L\Delta t \left(1 + \left(1 - \frac{\beta_2}{\lambda m_2} \right) + \dots + \left(1 - \frac{\beta_2}{\lambda m_2} \right)^{k-2} \right) \right) \\
 &+ \lim_{k \rightarrow \infty} \left(\left(1 - \frac{\beta_2}{\lambda m_2} \right)^{k-1} |e(1)| \right) \\
 &= \lim_{k \rightarrow \infty} \left(L\Delta t \frac{1 - \left(1 - \frac{\beta_2}{\lambda m_2} \right)^{k-1}}{1 - \left(1 - \frac{\beta_2}{\lambda m_2} \right)} \right) + 0 \\
 &= \frac{\lambda m_2 L \Delta t}{\beta_2}
 \end{aligned}$$

Fig. 6 The frames on the target

(44)

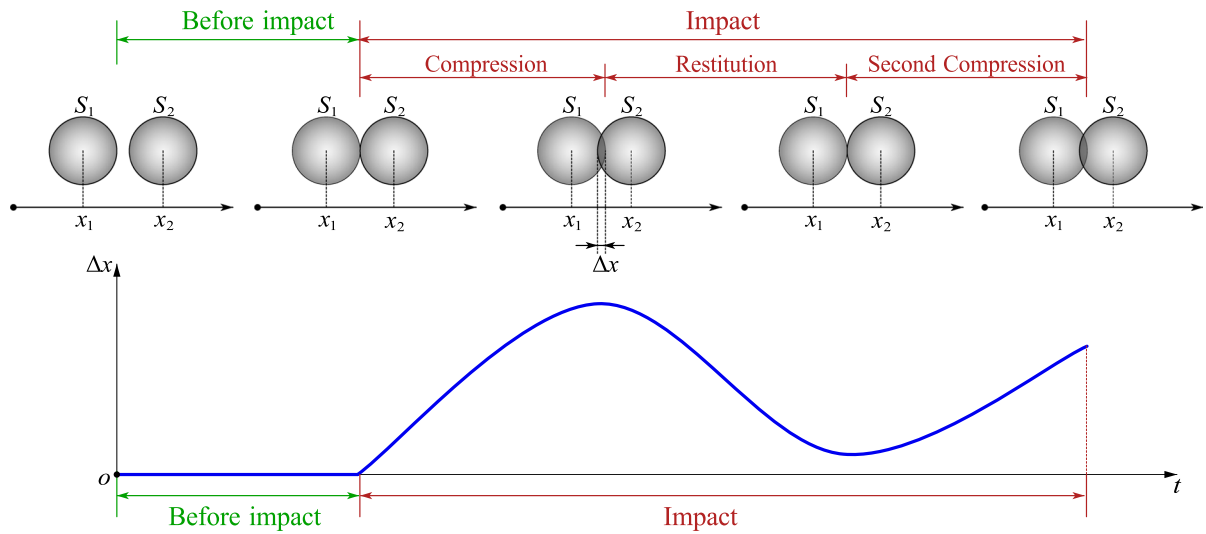


Fig. 7 The collision process with a secondary compression phase

Table 1 Selected contact models in this paper

Contact force model $F_c = k_c \Delta x^n + \chi \Delta \dot{x}^m \Delta \dot{x}$	Power exponent n	Power exponent m	Hysteresis damping factor χ	Applicable range of COR	COR values in this paper
LN model	3/2	3/2	$\chi = \frac{3k_c(1-e^2)}{4\Delta \dot{x}(0)}$	$e > 0.8$	0.9
Flores model	3/2	3/2	$\chi = \frac{8k_c(1-e)}{5e\Delta \dot{x}(0)}$	$0 \leq e \leq 1$	0.8
Gonthier model	3/2	3/2	$\chi \approx \frac{k_c(1-e^2)}{e\Delta \dot{x}(0)}$	$0 < e \leq 1$	0.6
GH model	3/2	3/2	$\chi = \frac{k_c}{e\Delta \dot{x}(0)}$	$e < 0.3$	0.2

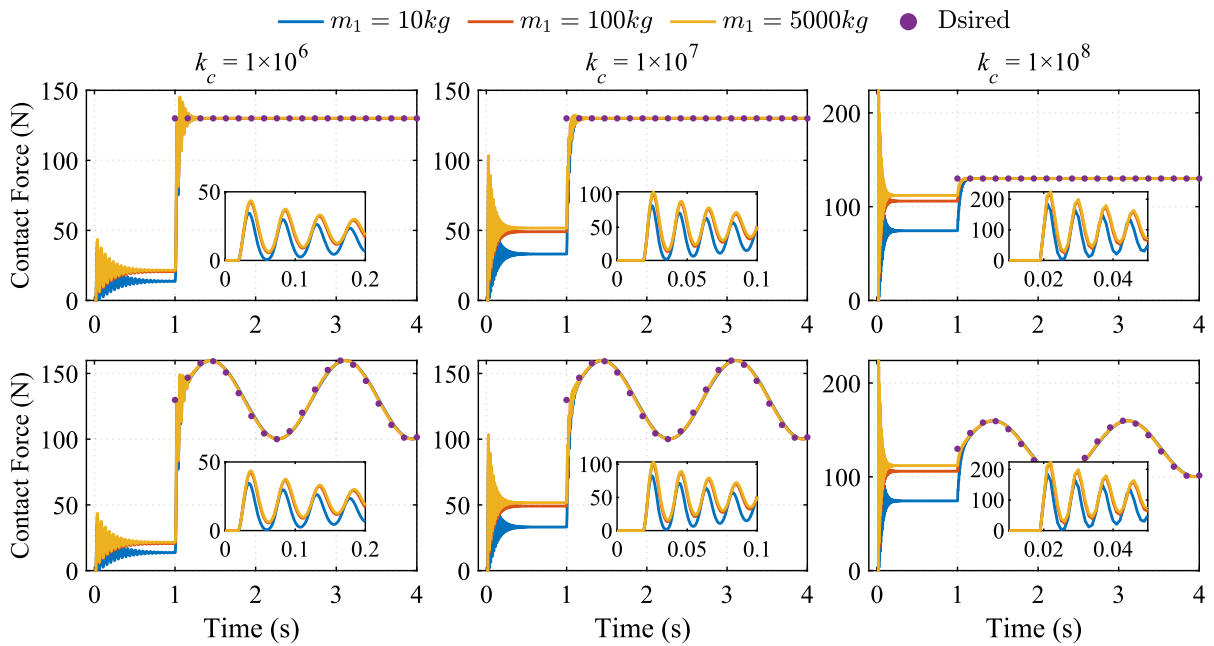


Fig. 8 The control results of contact force when using the LN model ($e = 0.9$)

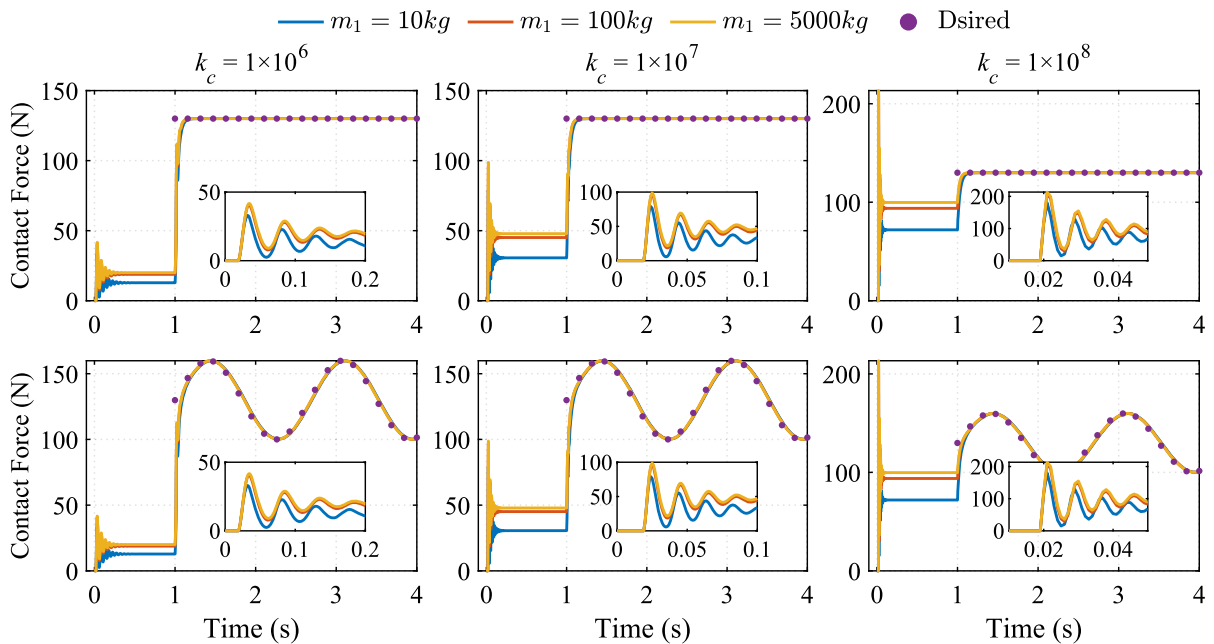


Fig. 9 The control results of contact force when using the Flores model ($e = 0.8$)

Therefore, we can obtain that the tracking error is bounded as $F_c^*(k)$ is time-varying. In summary, we have demonstrated ARC is effective for tracking the desired contact force.

3.3 The ARC-based contact control strategy

Although the previous sections demonstrate the effectiveness of the ARC method for maintaining contact and controlling contact force in the one-

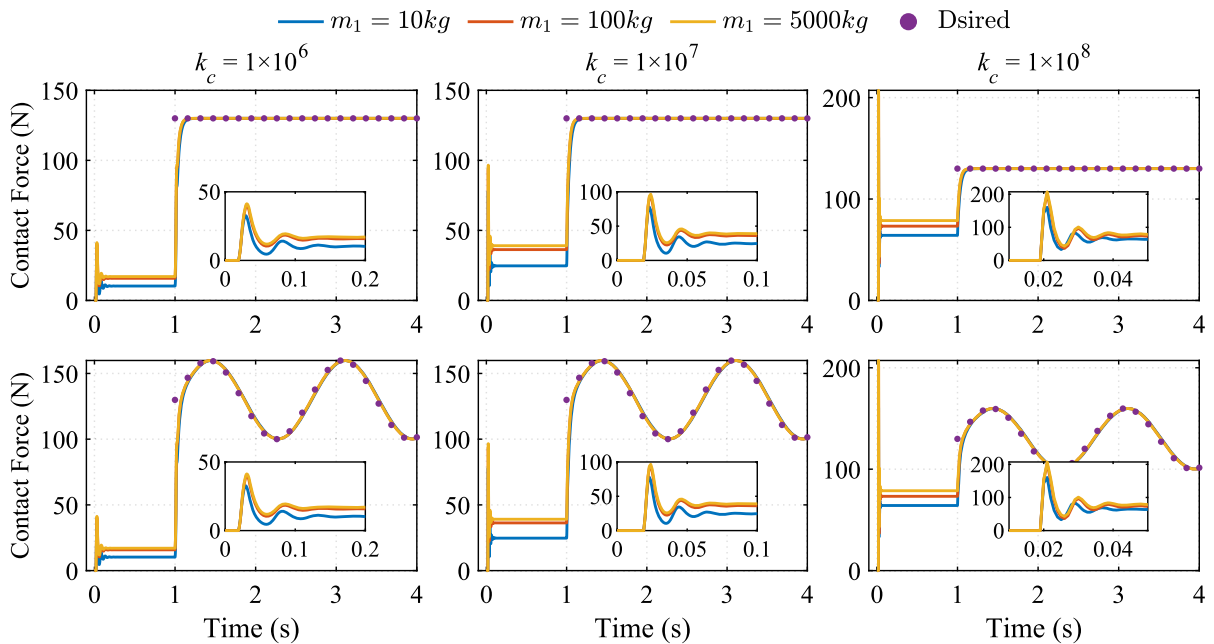


Fig. 10 The control results of contact force when using the Gonthier model ($e = 0.6$)

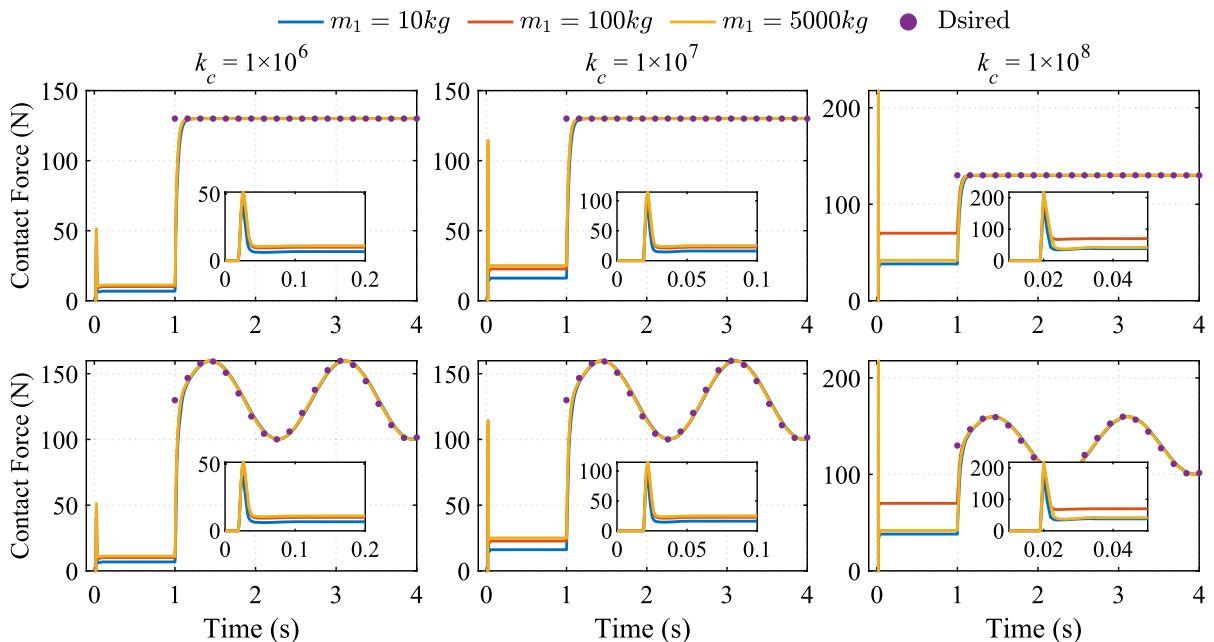


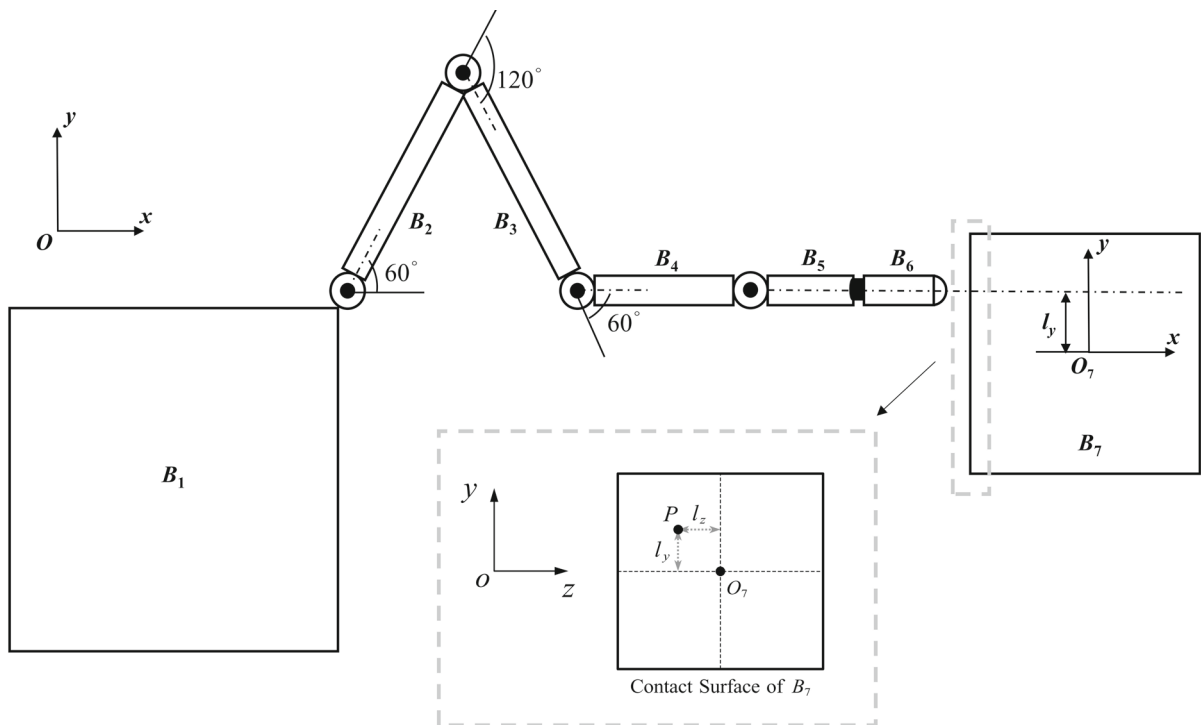
Fig. 11 The control results of contact force when using the GH model ($e = 0.2$)

dimension collision scenario, this method cannot be employed directly in the detumbling scenario. This is because the contact in the detumbling process is not a one-dimension scenario. To address this issue, we adopted a force/position hybrid control strategy, in

which the space robot has two controllers: a force controller and a motion controller. The former's objective is to control the relative movement between the contactor and the target along the contact force direction, and the latter aims to let the tip of the robotic

Table 2 Inertial parameters of the space robot and the non-cooperative target

Body	Mass (kg)	I_{xx} (kg·m ²)	I_{yy} (kg·m ²)	I_{zz} (kg·m ²)
B_1 (Base)	2200	370	370	370
B_2	4	6.5×10^{-4}	5×10^{-2}	5×10^{-2}
B_3	4	6.5×10^{-4}	5×10^{-2}	5×10^{-2}
B_4	3.5	4×10^{-4}	3.5×10^{-2}	3.5×10^{-2}
B_5	2.5	3.5×10^{-4}	3×10^{-2}	3×10^{-2}
B_6 (Contactor)	2.5	3.5×10^{-4}	3×10^{-2}	3×10^{-2}
B_7 (Target)	1000	200	100	150

**Fig. 12** Initial configurations of the space robot and non-cooperative satellite (Point P is the contact point)**Table 3** Initial linear and angular velocities of the target, and the position of the contact point

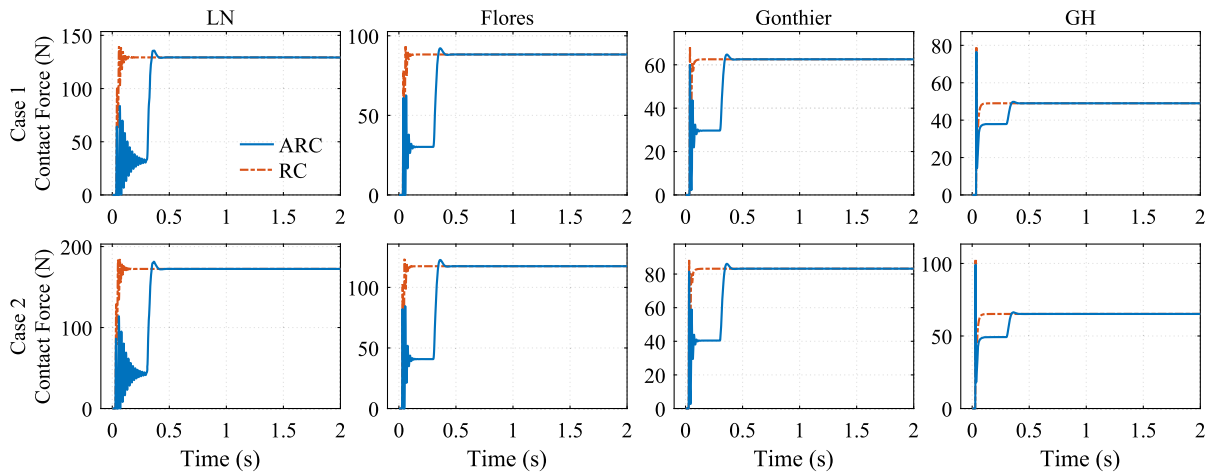
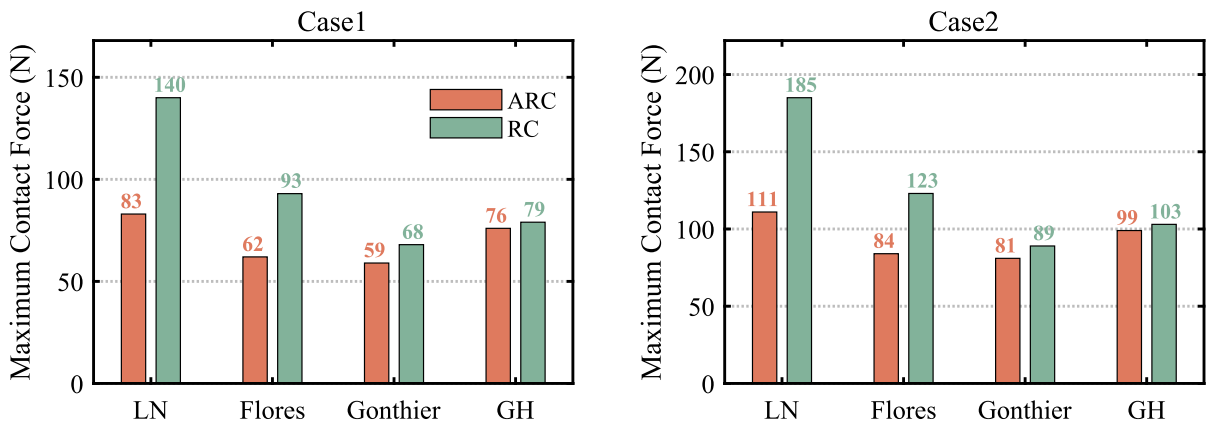
Case No	v_{target} (m/s)	ω_{target} (rad/s)	Position of contact point P (m)
Case 1	$[-0.05, 0.08, -0.02]^T$	$[0.3, 0.1, 0.5]^T$	$l_y = 0.06, l_z = 0.04$
Case 2	$[-0.08, 0.07, 0.06]^T$	$[-0.4, 0.4, 0.2]^T$	$l_y = 0.06, l_z = 0.04$

arm (body B_5), as shown in Fig. 2, track the desired trajectory in the other directions. In this study, ARC is utilized in designing the force controller, and the control force is generated by a linear motor at joint H_7 , as shown in Fig. 2. The motion controller is designed by the computed torque method. Since the computed

torque method is a classic control method, this study will not present the details of the motion control. Next, this subsection will present the desired trajectory of the body B_5 and the desired joint trajectory of the robotic arm.

Table 4 Joint motor parameter settings in simulation

Joint	Torque/force limits	Electromagnetic time τ	Gain coefficient K_g
H_1 - H_6 (Rotary)	150 N·m	12 ms	83
H_7 (Linea)	200 N	10 ms	100

**Fig. 13** Contact forces between the contactor and target surface under different contact models in Cases 1 and 2**Fig. 14** Comparison of maximum contact force in the initial stage of the collision ($t \leq 0.3$ s) between ARC-based and RC-based contact control strategy

As illustrated in Fig. 6, the inertial reference frame $\sum$ is defined along the orbit of the target, and the frame $\sum_s$ represents a body-fixed frame located on the contact surface A . At moment τ , the desired position of body B_5 in the frame $\sum$ is expressed as:

$$\mathbf{r}_{B_5}^d(\tau) = \boldsymbol{\rho}_{B_5}(\tau) + \mathbf{r}_{O_s}(\tau) = \mathbf{A}_{surface}(\tau) \boldsymbol{\rho}'_{B_5}(\tau) + \mathbf{r}_{O_s}(\tau) \quad (45)$$

where $\boldsymbol{\rho}'_{B_5}(\tau) = [\Delta x_s(\tau) \Delta y_s(\tau) \Delta z_s(\tau)]^T$ represents the projection of the vector from the mass center of body B_5 to $\sum_s$ in the frame $\sum_s$. By differentiating the

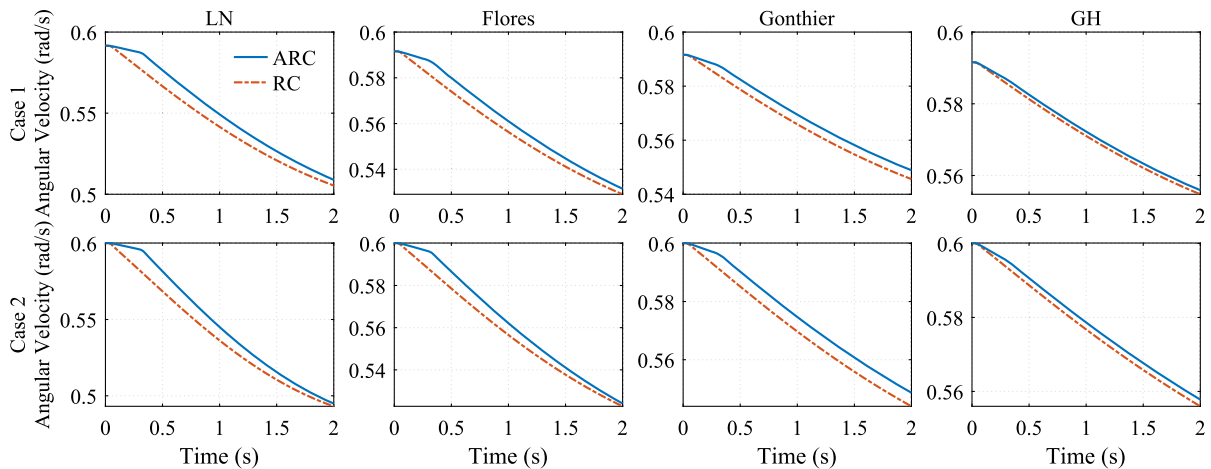


Fig. 15 The magnitude of the resultant angular velocity vector of the target with respect to the base

above equation, the velocity of the body B_5 can be expressed as:

$$\dot{\mathbf{r}}_{B_5}^d(\tau) = \overset{\circ}{\rho}_{B_5} + \boldsymbol{\omega}_{surface}(\tau) \times \rho_{B_5}(\tau) + \dot{\mathbf{r}}_{O_s}(\tau) \quad (46)$$

where $\overset{\circ}{\rho}_{B_5}(\tau) = \mathbf{A}_{surface}(\tau)[\Delta\dot{x}_s(\tau) \ \Delta\dot{y}_s(\tau) \ \Delta\dot{z}_s(\tau)]^T$, and $\boldsymbol{\omega}_{surface}(\tau)$ denotes the angular velocity of the frame $\sum_s$ with respect to the inertial reference frame $\sum$. Further differentiating the velocity (46) yields the acceleration:

$$\begin{aligned} \ddot{\mathbf{r}}_{B_5}^d(\tau) = & \overset{\circ\circ}{\rho}_{B_5}(\tau) + \dot{\omega}_{surface}(\tau) \times \rho_{B_5}(\tau) + 2\omega_{surface}(\tau) \\ & \times \overset{\circ}{\rho}_{B_5}(\tau) + \omega_{surface}(\tau) \times \omega_{surface}(\tau) \times \rho_{B_5}(\tau) \\ & + \ddot{\mathbf{r}}_{O_s}(\tau) \end{aligned} \quad (47)$$

where $\overset{\circ\circ}{\rho}_{B_5}(\tau) = \mathbf{A}_{surface}(\tau)[\Delta\ddot{x}_s(\tau) \ \Delta\ddot{y}_s(\tau) \ \Delta\ddot{z}_s(\tau)]^T$, and $\dot{\omega}_{surface}(\tau)$ denotes the angular acceleration of the frame $\sum_s$ with respect to the inertial reference frame $\sum$.

At time τ , the desired attitude $\boldsymbol{\theta}_{B_5}^d(\tau)$, angular velocity $\boldsymbol{\omega}_{B_5}^d(\tau)$, and angular acceleration $\dot{\omega}_{B_5}^d(\tau)$ of the body B_5 are equal to those of the surface A (Fig. 6), denoted as $\boldsymbol{\theta}_{surface}(\tau)$, $\boldsymbol{\omega}_{surface}(\tau)$, and $\dot{\omega}_{surface}(\tau)$. These values can be accurately obtained using lasers or cameras. Consequently, the desired motion of the body B_5 at moment τ can be expressed as:

$$\begin{cases} \mathbf{q}_{B_5}^d(\tau) = [\mathbf{r}_{B_5}^d(\tau); \theta_{B_5}^d(\tau)] \\ \dot{\mathbf{q}}_{B_5}^d(\tau) = [\dot{\mathbf{r}}_{B_5}^d(\tau); \dot{\omega}_{B_5}^d(\tau)] \\ \ddot{\mathbf{q}}_{B_5}^d(\tau) = [\ddot{\mathbf{r}}_{B_5}^d(\tau); \ddot{\omega}_{B_5}^d(\tau)] \end{cases} \quad (48)$$

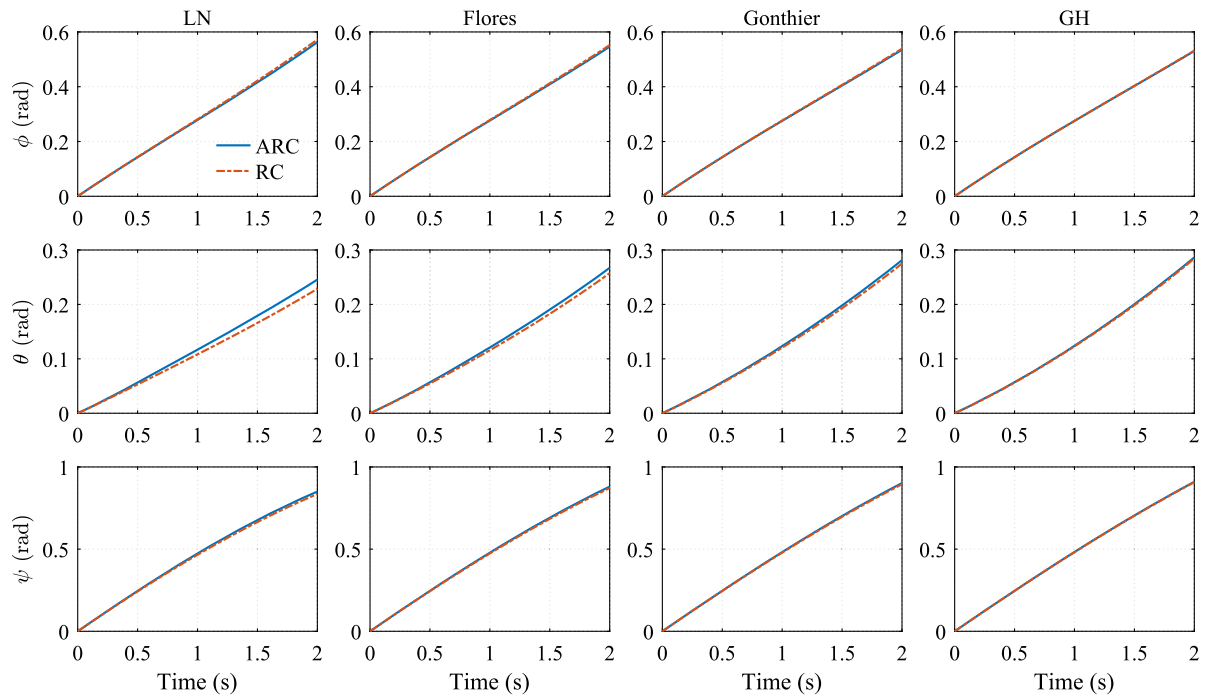
Throughout the detumbling process, the base of the space robot maintains a constant motion state within the inertial reference frame $\sum$. Leveraging the inverse kinematics of the space robot, the desired configuration parameters $\mathbf{q}_{arm}^d(\tau)$, $\dot{\mathbf{q}}_{arm}^d(\tau)$, and $\ddot{\mathbf{q}}_{arm}^d(\tau)$ can be directly calculated.

4 Numerical simulations

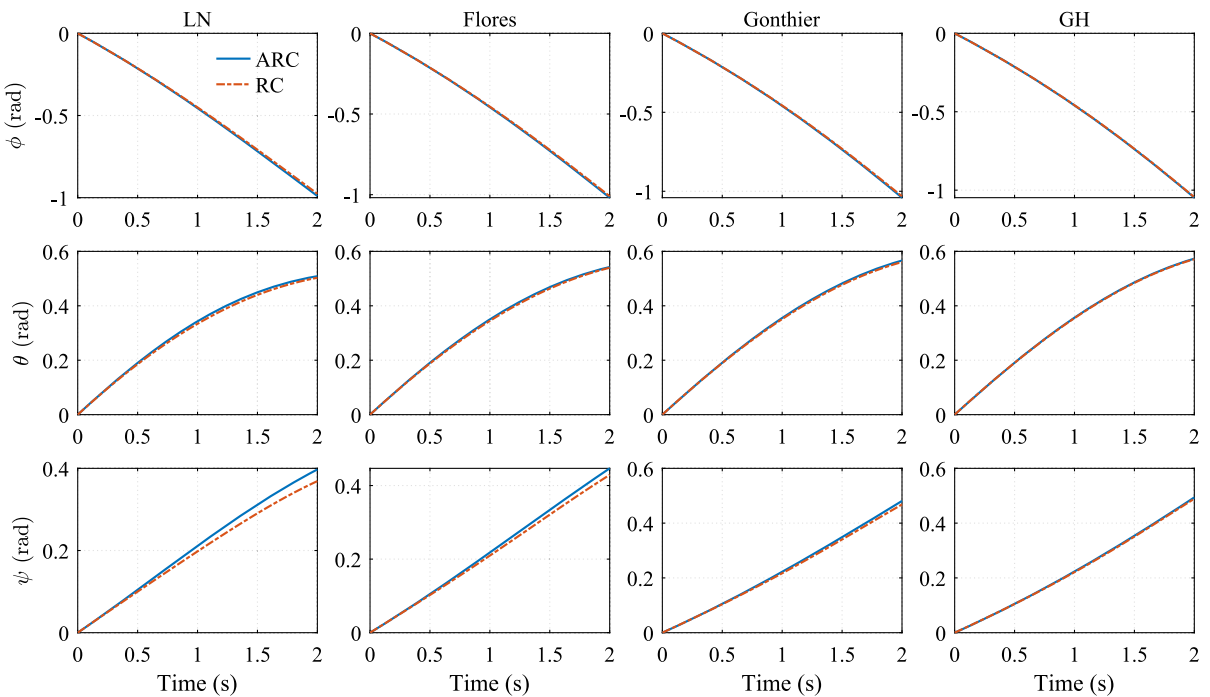
This section first verifies the effectiveness of ARC in the two-ball contact model, and then evaluates the effectiveness of the presented ARC-based contact control strategy and compares its performance with the resistance control (RC) [52]. Finally, the contact control strategy's robustness against measurement noise is assessed.

4.1 Control results of ARC in two-ball collision model

In this section, the scenario of the two-ball collision, as depicted in Fig. 3, is used to validate ARC's effectiveness in preventing separation and tracking the desired contact force. Note that the elastoplastic contact model was not employed in our simulations.



(a)



(b)

Fig. 16 The XYZ Euler angles of the target with respect to the base under different contact models: **a** Case 1 and **b** Case 2

Table 5 The constant desired contact force settings for ARC-based contact control strategy

	LN	Flores	Gonthier	GH
Case 1	129 N	88 N	63 N	49 N
Case 2	172 N	118 N	83 N	65 N

The reasons are as follows: when the collision involves ARC, the collision bodies will not immediately separate after contact. The collision process will include a secondary compression phase, as shown in Fig. 7. Since most elastoplastic contact models are incapable of describing the contact force during the secondary compression phase after collision, conducting control simulations using these models is not feasible. So, this section only presents the control simulation results with different viscoelastic contact models.

4.1.1 Simulation settings

In this section, to demonstrate the effectiveness of ARC does not depend on a specific contact model, the following four viscoelastic contact models are selected to describe the contact force: the Lanakrani-Nikraves (LN) model [62], Flores model [63], Gonthier model [64], and Gharib-Hurmuzlu (GH) model [65]. As shown in Table 1, we set different coefficient of restitution (COR) values for each contact model based on its applicable range. The mass of S_2 is fixed at $m_2 = 2.5$ kg, while the mass of S_1 is varied across three cases: $m_1 = 10$ kg, $m_1 = 100$ kg, and $m_1 = 5000$ kg. The initial relative velocity is 0.1 m/s. The control parameters of ARC are $\beta_1 = 0.5$ and $\beta_2 = 0.02$. The sampling time of control is 1 ms. During the contact force tracking phase, the desired contact force $F_c^*(t)$ is defined as:

$$F_c^*(t) = \begin{cases} 130 & \text{if } F_c^*(t) \text{ is constant} \\ 130 + 30 \sin(1.2\pi(t - t_s)) & \text{if } F_c^*(t) \text{ is time - varying} \end{cases} \quad (49)$$

where t_s is the switching moment between the two control phases, which is set to $t_s = 1$ s in this section.

4.1.2 Effectiveness of ARC for different viscoelastic contact models

The simulation results of ARC in the scenario of the two-ball contact are shown in Figs. 8, 9, 10, 11. In all cases, the contact force initially exhibits oscillations and then stabilizes to a positive constant value in the first control phase ($0 \leq t \leq 1$ s). Those facts illustrate that ARC effectively prevents separation and forms stable contact. Additionally, once the control process entered the contact force tracking phase ($t > 1$ s), the ARC demonstrated good tracking performance for both constant and time-varying desired contact forces. Therefore, ARC can achieve control objectives in the two-ball contact model. It is also worth noting that ARC achieves the desired control objectives under all four viscoelastic contact models considered in this study. This indicates that the effectiveness of ARC does not rely on any specific contact model, thereby demonstrating its strong robustness to contact model variations.

4.2 Control results of the ARC-based contact control strategy in detumbling task

This subsection first evaluates the effectiveness of the ARC-based contact control strategy by simulations. Subsequently, an evaluation of the robustness of the proposed contact control strategy against measurement noise will be conducted.

4.2.1 Simulation settings

The structural configuration of the space robot and the non-cooperative target is depicted in Fig. 2. The system includes the base (B_1), the robotic arm ($B_2 \sim B_5$), the contactor (B_6) and the target (B_7), whose inertial properties are listed in Table 2. The lengths of the bodies B_2 to B_6 are 0.84 m, 0.84 m, 0.4 m, 0.2 m, and 0.2 m, respectively. The base (B_1) and the target (B_7) are cubes with side lengths of 1 m and 0.5 m, respectively. The initial configuration of the space robot and the target are illustrated in Fig. 12, while Table 3 details the initial velocity and angular velocity of the target and the position of the contact point. The control system operates with a sampling

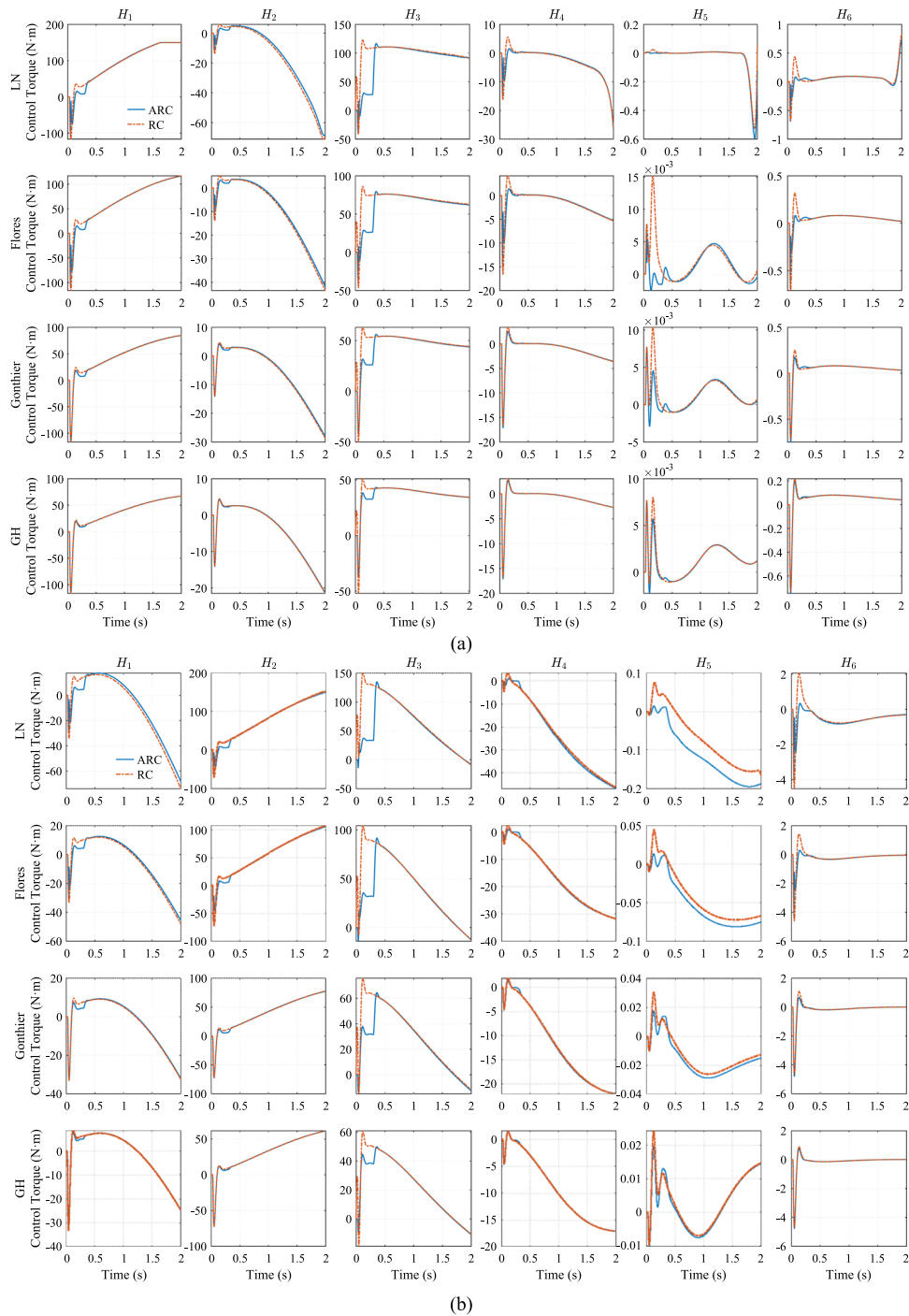


Fig. 17 Control torques of the robotic arm's joint H_1 – H_6 under different contact models: **a** Case 1 and **b** Case 2

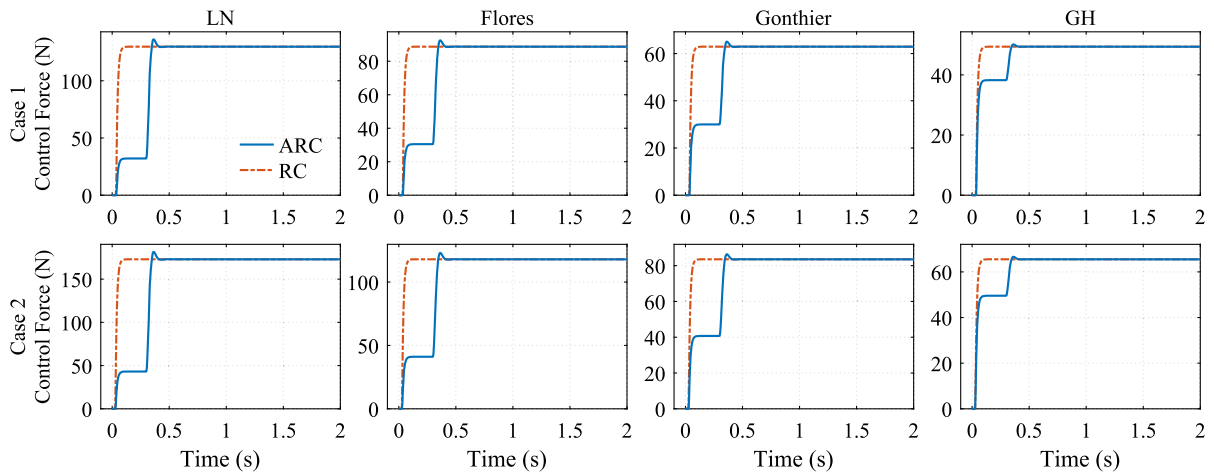


Fig. 18 Control forces of the robotic arm's translation joint H_7 in Cases 1 and 2

time of 1 ms. For the ARC method, the control parameters are $\beta_1 = 0.5$ and $\beta_2 = 0.05$. For the RC method, the control parameter is $\beta = \beta_s$ (the definitions of β and β_s are in accordance with Ref. [52]).

To simulate the dynamic response of the joint actuators, the motors installed in the joints of the space robot arm are modeled using a first-order approximation. This approximation can be expressed by the following equation:

$$\dot{T} = K_g(\tilde{T} - T) \quad (50)$$

where T represents the actual output torque/force, $\tilde{T}$ is the expected output torque/force, and K_g denotes the gain coefficient, calculated based on the electromagnetic time constant of the motor. In addition, saturation limits on joint torque/force outputs are considered. The specific actuator parameters for the joint motors are provided in Table 4. Other non-ideal joint effects such as joint friction, time delays, and velocity and acceleration limits are not considered in this study.

4.2.2 Effectiveness of the ARC-based contact control strategy in detumbling task

In this section, the four contact models in Table 1 were employed to describe the contact force, with a stiffness of $k_c = 1 \times 10^7 \text{ N/m}^{2/3}$. The simulation results in the two cases (Table 3) are shown in Figs. 13, 14, 15, 16. Figure 13 shows the contact force between the

contactor and the target surface under the ARC-based and RC-based contact control strategies. Both strategies are capable of preventing post-collision separation. However, as shown in Fig. 14, the ARC-based strategy exhibits a smaller maximum contact force during the initial stage of the collision ($t \leq 0.3 \text{ s}$), particularly under the LN and Flores models. This indicates that the ARC-based strategy can mitigate the negative impact of collisions on the robotic arm and the target. Although the RC-based strategy can produce larger contact forces after stable contact is established ($t > 0.3 \text{ s}$), the ARC-based strategy is capable of adjusting the contact force to match that of the RC-based strategy through contact force tracking control, as shown in Fig. 13. The desired contact force values are listed in Table 5. Therefore, as shown in Figs. 15 and 16, the ARC-based strategy can achieve a detumbling effect that approximates that of the RC-based strategy.

Figures 17 and 18 depict the control torques (H_1 – H_6) and control forces (H) exerted on the robotic arm's joints under the ARC-based and RC-based contact control strategies. It is evident that the joint output under both strategies remains within the actuator limits, specifically a torque limit of $\tau \leq 150 \text{ N}\cdot\text{m}$ for joints H_1 – H_6 and a force limit of $f \leq 200 \text{ N}$ for joint H_7 . During the initial phase (0–0.3 s), sharp variations in control signals are observed in all cases due to collision impacts. Notably, compared to RC, the ARC-

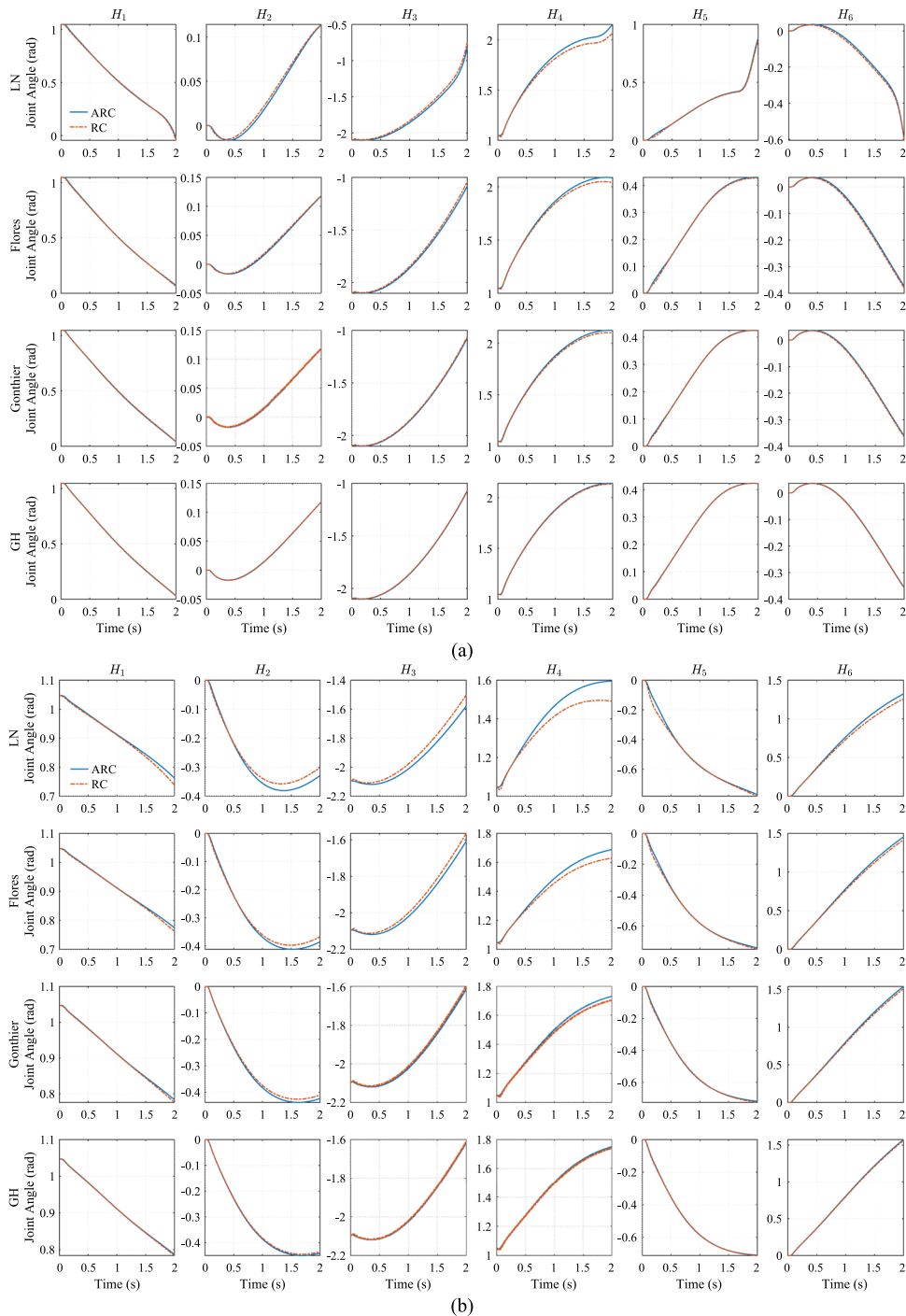


Fig. 19 Joint angles of the robotic arm's joint H_1 – H_6 under different contact models: **a** Case 1 and **b** Case 2

based strategy yields smaller peak control efforts for most joints, thereby reducing actuator load.

Figures 19 and 20 illustrate the evolution of the robotic arm's joint under different control strategies.

As shown in Fig. 19, the rotation joint trajectories (H_1 – H_6) under both ARC-based and RC-based strategies are smooth, indicating that the system does not experience severe vibrations. However, for the

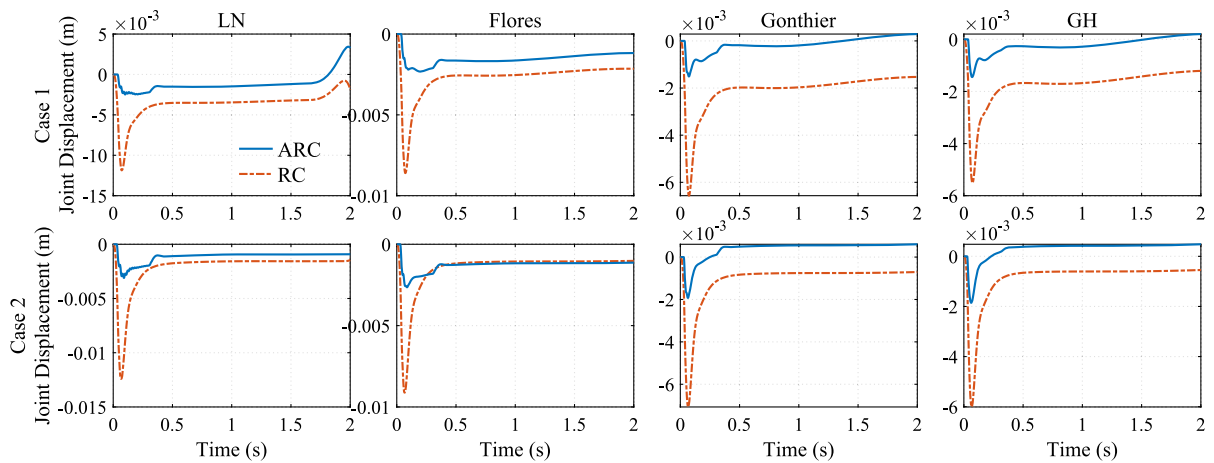


Fig. 20 Joint displacement of the robotic arm's translation joint H_7 in Cases 1 and 2

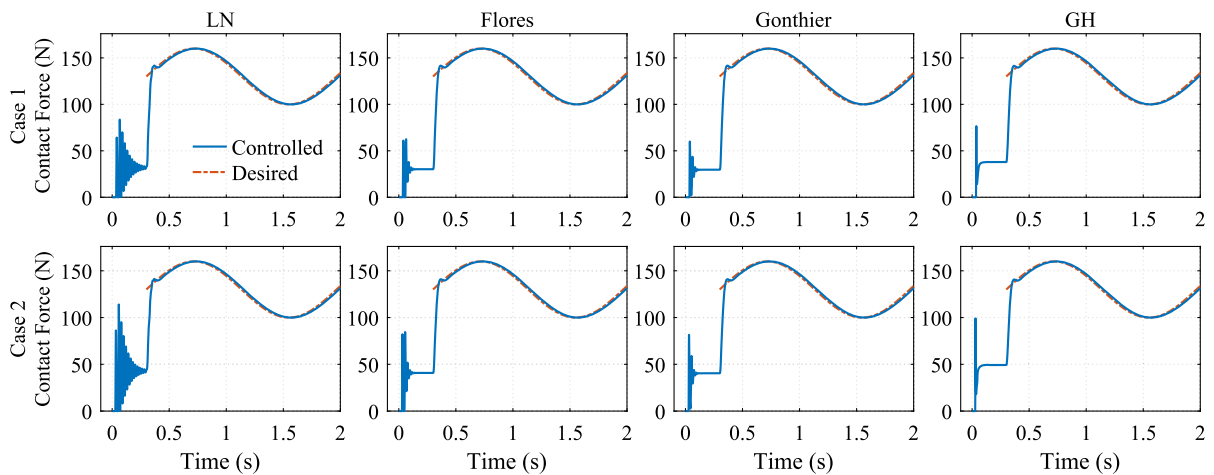


Fig. 21 Tracking performance of controlled contact force under a time-varying desired contact force

Table 6 The measurement errors reported in previous studies for each relevant measurand

Measurand (Quantity being measured)	Measurement error	References
Target's position	< 0.005 m	[66]
Target's velocity	0.00658 m/s	[67]
Target's attitude (XYZ Euler Angle)	< 0.0175 rad	[66]
Target's angular velocity	0.023 rad/s	[67]
Contact force	< 1 N	[68]

translational joint (Fig. 20), the RC-based trajectory exhibits more pronounced variations at the initial stage, with higher peak values. This imposes greater demands on the actuator and may affect control efficiency.

Based on the above results, it can be concluded that both ARC-based and RC-based strategies are capable

of achieving the control objective of preventing post-collision separation. However, compared to RC, the ARC-based strategy achieves this goal with a smaller maximum contact force and reduced control effort. Therefore, ARC-based strategy demonstrates better overall performance.

Table 7 Gaussian noise parameter settings for each measurand in the simulation

Measurand (Quantity being measured)	Standard deviation (σ)	Mean (μ)
Target's position	0.1 m	0
Target's velocity	0.05 m/s	0
Target's attitude (XYZ Euler Angle)	0.02 rad	0
Target's angular velocity	0.1 rad/s	0
Contact force	1 N	0

In addition to preventing post-collision separation, the ARC-based contact control strategy is also capable of tracking a desired contact force. In Fig. 13, we have demonstrated the ability of the ARC-based contact control strategy to track a constant desired contact force. Considering that in the detumbling task, the desired contact force required may also be time-varying. In the following, we further validate the effectiveness of the ARC-based strategy in tracking the time-varying contact force. In this subsection, the desired time-varying contact force is defined as:

$$F_c^*(t) = 130 + 30 \sin(1.2\pi(t - t_s)) \quad (51)$$

where the switch time t_s is set to 0.3s. The simulation results are shown in Fig. 21. As shown in this figure, in the contact force tracking phase ($t > 0.3$ s), the ARC-based strategy demonstrates good performance in tracking the time-varying contact force. This offers significant flexibility for future designs of the optimal detumbling strategies. Therefore, the ARC-based contact control strategy has great potential for detumbling applications.

4.2.3 Robustness of the ARC-based contact control strategy against measurement noise

The ARC-based contact control strategy proposed in this study requires measurements of target motion and contact force. In the detumbling mission, measurement noise of these physical quantities is difficult to avoid. Whether the ARC-based strategy exhibits robustness against these noises is a problem worth studying. This subsection introduces Gaussian noise into the measurement data of the target motion and the contact force. To determine reliable noise parameters, we surveyed relevant literature. Table 6 summarizes the measurement errors reported in previous studies

for each relevant measurand. Based on these values, we defined the Gaussian noise parameters used in our simulation (Table 7). A comparison between Tables 6 and 7 reveals that the standard deviations of the noise adopted in our study are significantly larger than the reported measurement errors. This setting enables a more rigorous evaluation of the proposed strategy's robustness against measurement noise.

In addition, to attenuate the impact of noise, a first-order filter is utilized on the measured data of the target motion, which is expressed by the following equation:

$$X_k = \eta Y_k + (1 - \eta)X_{k-1} \quad (52)$$

where Y_k represents the sample value at time k , X_{k-1} and X_k are the filtered value at time $k-1$ and k , respectively, and η is the filter coefficient. In this study, η is set to 0.1. Besides, the desired contact force $F_c^*(t)$ in this subsection is defined as:

$$F_c^*(t) = \begin{cases} 80 & \text{if } F_c^*(t) \text{ is constant} \\ 80 + 20 \sin(1.2\pi(t - t_s)) & \text{if } F_c^*(t) \text{ is time-varying} \end{cases} \quad (53)$$

where the switch time t_s is set to 0.3s. The other simulation settings are the same as those in Sect. 4.2.2.

The simulation results are shown in Figs. 22, 23, 24, 25, 26. First, Fig. 22 shows the effect of the introduction of measurement noise on target motion and contact force measurement information (since the noise settings are consistent across all cases, only one case is presented here as an example). The added noise generates an error between the measured and true values, indicating that the noise has a significant impact on measurement accuracy. On this basis, Fig. 23 shows the performance of the ARC-based strategy in controlling the contact force under

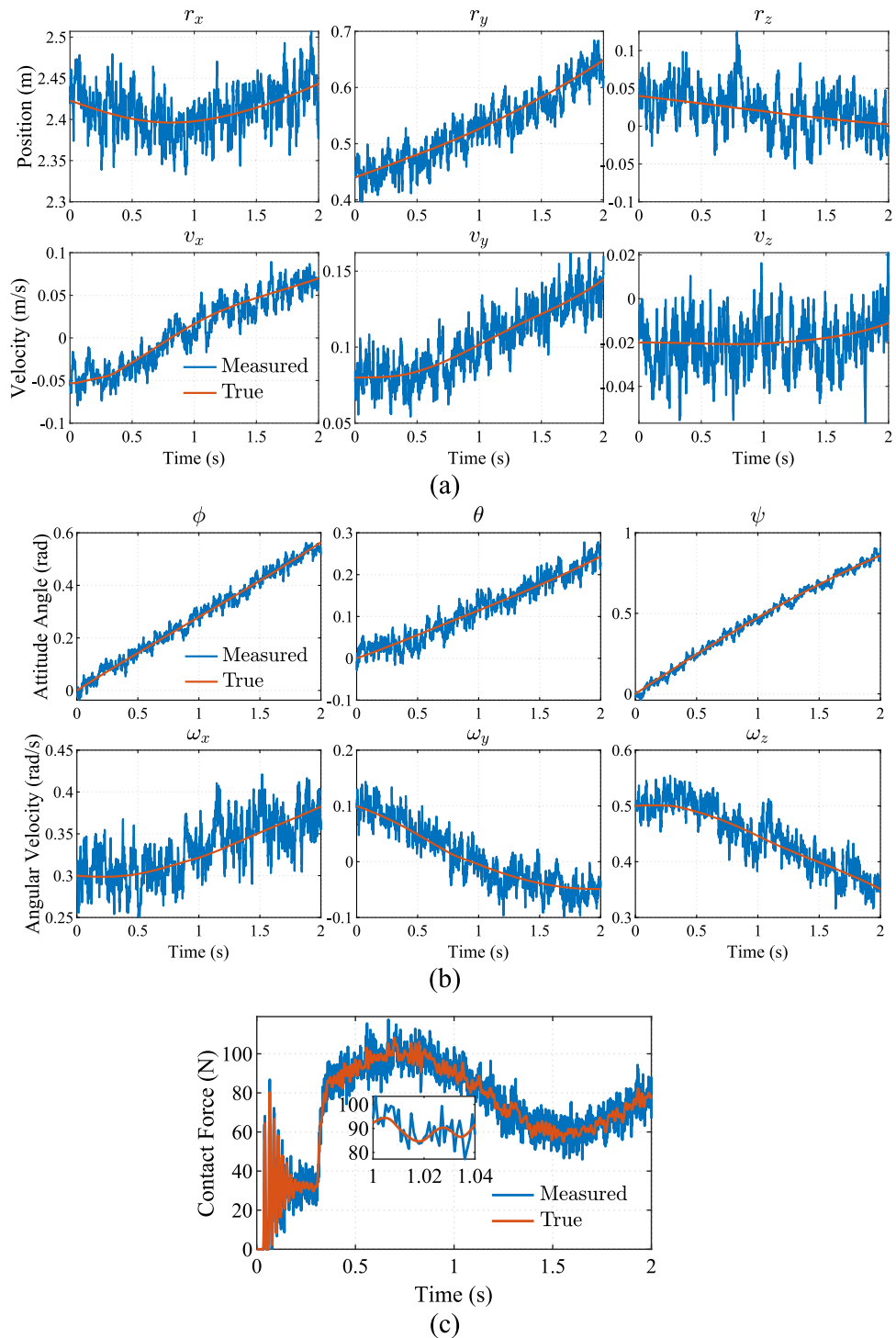


Fig. 22 Measured and true data of the target motion and the contact force under LN contact model in Case 1: **a** target position and velocity; **b** target attitude and angular velocity; **c** contact force

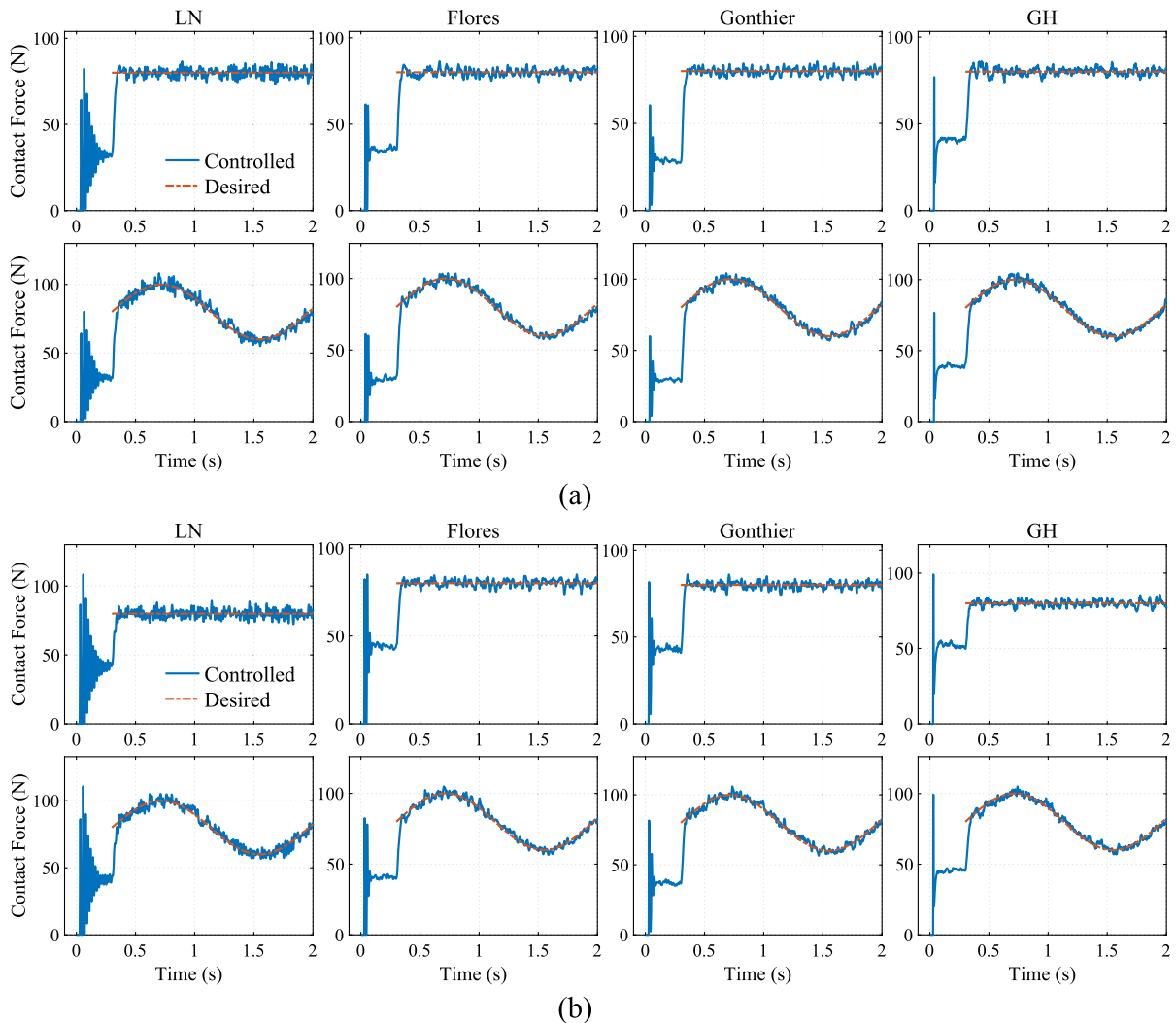


Fig. 23 The controlled and desired contact forces under different contact models in **a** Case 1 and **b** Case 2

measurement noise across all cases. Although the presence of noise introduces fluctuations in the contact force, the proposed strategy still effectively prevents post-collision separation and tracks the desired contact force. In addition, Fig. 24 depicts the target's resultant angular velocity with respect to the base under noise and noise-free conditions. The resultant angular velocity of the target decreases steadily over time even with measurement noise present, exhibiting a similarly decreasing trend with the noise-free cases. This indicates that the measurement noise does not significantly affect the detumbling effect of the proposed strategy. Furthermore, the results in Figs. 25

and 26 show that the joint torques/forces do not exceed the predefined saturation limits: 150 N·m for H_1 - H_6 (rotational) and 200 N for H_7 (translational). Overall, these results demonstrate the robustness of the proposed ARC-based contact control strategy against the measurement noise of target motion and contact force.

5 Conclusion

This study proposes a novel contact control approach, termed ARC, to prevent the separation

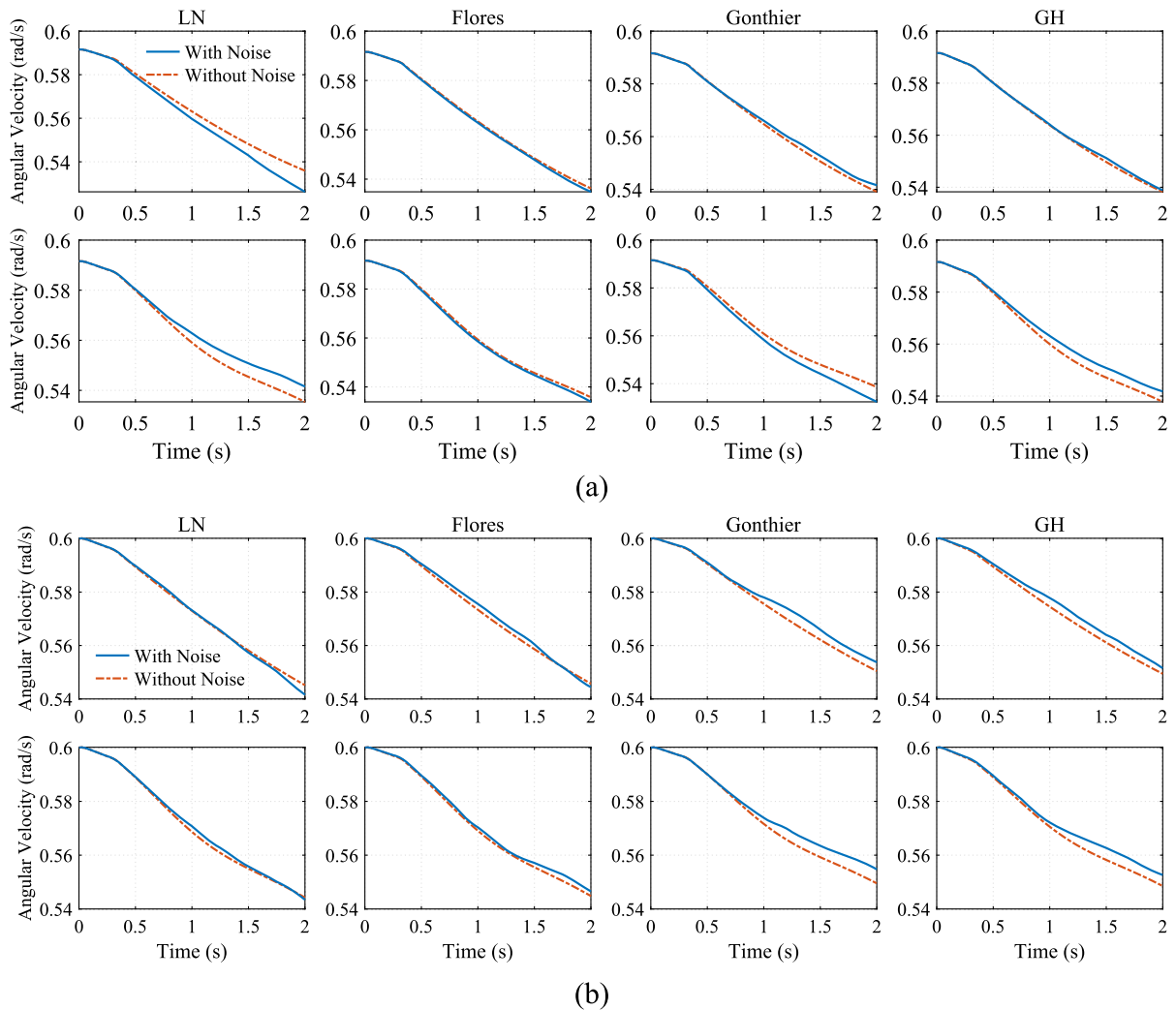


Fig. 24 The magnitude of the target's resultant angular velocity vector with respect to the base under different contact models in **a** Case 1 and **b** Case 2. The first and second rows represent results for constant and time-varying desired contact forces, respectively

of colliding objects and track the desired contact force. Notably, this method does not require prior knowledge of the target's inertial and contact parameters, eliminating the need for parameter identification. Theoretical analysis illustrates that ARC's effectiveness does not depend on the specific contact model. To apply this method to the detumbling tasks, this study introduces an ARC-based contact control strategy. In the simulation section, a

two-ball model was used to verify the capability of the ARC method in preventing separation and tracking desired contact forces. Subsequently, the ARC-based contact control strategy was validated in the detumbling tasks. The simulation results revealed that this contact control strategy effectively achieved the desired control objectives and outperformed the RC-based strategy. Furthermore, the robustness of the ARC-based contact control

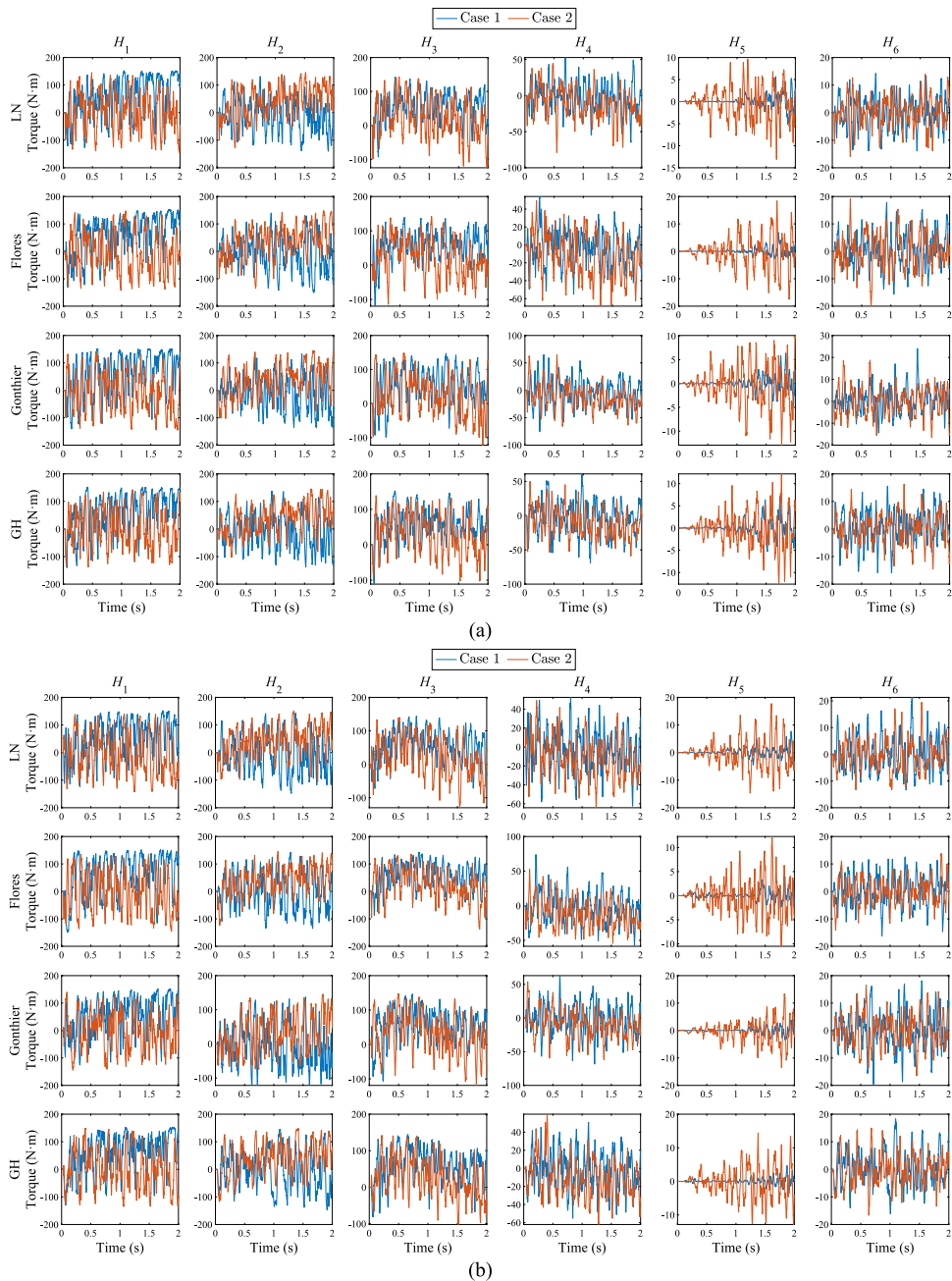


Fig. 25 Control torques of the robotic arm's joint H_1 – H_6 under different contact models in Cases 1 and 2: **a** constant desired contact force; **b** time-varying desired contact force

strategy against measurement noise was validated through numerical simulations. These findings suggest that the ARC method can serve as an efficient contact control approach for detumbling non-

cooperative targets, thereby expanding the potential application of robotic arms in this field.

Future research directions include exploring optimal detumbling strategies based on the ARC method

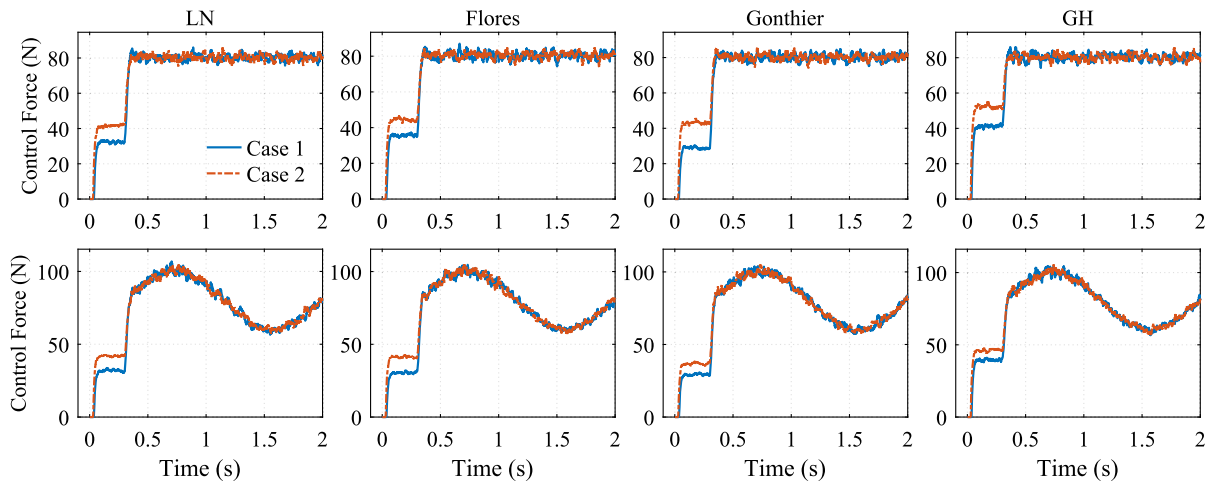


Fig. 26 Control forces of the robotic arm's translation joint H_7 under different contact models in Cases 1 and 2. The first and second rows represent results for constant and time-varying desired contact forces, respectively

and conducting experimental validation of the proposed approach.

Author contributions Ruhao Wang wrote the main manuscript text and the source code. Xiaofeng Liu wrote and reviewed the manuscript. Guoping Cai reviewed the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval This declaration is not applicable.

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